

Python 3.14.7 (tags/v3.14.7:823f032, Aug 5 2026, 10:51:32) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:\Users\Rahul\Documents\CSE1104_Deep_learning\Assignment
2\assignment2.py

=====
Epoch : 1, learning_rate : 0.0

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new

-> [Cycle Detected] Period: 1 updates

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 2, learning_rate : 0.0

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 3, learning_rate : 0.0

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 4, learning_rate : 0.0

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new

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-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 5, learning_rate : 0.0

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=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 6, learning_rate : 0.0

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=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 7, learning_rate : 0.0

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=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 8, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 9, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 10, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 11, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 12, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 13, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 14, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 15, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 16, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 17, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 18, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 19, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 20, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 21, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 22, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 23, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 24, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 25, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 26, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 27, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
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0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 28, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 29, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 30, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 31, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 32, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 33, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 34, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 35, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 36, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 37, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 38, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 39, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 40, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 41, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 42, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 43, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 44, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 45, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 46, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```

.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 47, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 48, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 49, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 50, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50

```

```

.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 51, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 52, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 53, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 54, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 55, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 56, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 57, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 58, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 59, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 60, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 61, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

=====
Epoch : 62, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 63, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 64, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 65, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 66, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 67, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 68, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 69, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 70, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 71, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 72, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 73, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 74, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 75, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 76, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 77, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 78, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 79, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 80, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 81, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 82, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 83, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 84, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 85, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 86, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 87, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 88, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 89, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 90, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 91, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 92, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 93, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 94, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 95, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 96, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```

.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 97, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 98, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 99, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 100, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50

```

```

.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 101, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 102, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 103, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 104, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 105, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 106, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 107, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 108, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 109, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 110, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 111, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

=====
Epoch : 112, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 113, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 114, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 115, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 116, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 117, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 118, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 119, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 120, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 121, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 122, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 123, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 124, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 125, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 126, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 127, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 128, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 129, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 130, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 131, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 132, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 133, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 134, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 135, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 136, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 137, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 138, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 139, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 140, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 141, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 142, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 143, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 144, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 145, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 146, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```

.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 147, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 148, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 149, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 150, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50

```

```

.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 151, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 152, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 153, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 154, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 155, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 156, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 157, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 158, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 159, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 160, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 161, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 0 | 0    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

=====
Epoch : 162, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 163, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 164, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 165, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 166, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 167, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 168, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 169, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		0		0		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 170, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 171, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 172, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 173, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 174, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 175, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 176, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 177, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 178, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 179, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 180, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 181, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 182, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 183, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 184, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 185, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 186, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 187, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 188, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	0	0	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 189, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 0  | 0    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 190, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 0  | 0    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 191, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 0  | 0    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 192, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 0  | 0    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 193, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 194, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 195, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 196, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 197, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 198, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 199, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 200, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
```

```

.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 1, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.40
0	1	0	0	0	[1.00, -1.00]	0.40	[1.00, -1.00]	0.40
1	0	0	1	-1	[1.00, -1.00]	0.40	[0.90, -1.00]	0.30
1	1	1	1	0	[0.90, -1.00]	0.30	[0.90, -1.00]	0.30

=====
Epoch : 2, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[0.90, -1.00]	0.30	[0.90, -1.00]	0.20
0	1	0	0	0	[0.90, -1.00]	0.20	[0.90, -1.00]	0.20
1	0	0	1	-1	[0.90, -1.00]	0.20	[0.80, -1.00]	0.10
1	1	1	0	1	[0.80, -1.00]	0.10	[0.90, -0.90]	0.20

=====
Epoch : 3, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[0.90, -0.90]	0.20	[0.90, -0.90]	0.10
0	1	0	0	0	[0.90, -0.90]	0.10	[0.90, -0.90]	0.10
1	0	0	1	-1	[0.90, -0.90]	0.10	[0.80, -0.90]	0.00
1	1	1	0	1	[0.80, -0.90]	0.00	[0.90, -0.80]	0.10

=====
Epoch : 4, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [0.90, -0.80] | 0.10 | [0.90, -0.80] | 0.00
.....
0 | 1 | 0 | 0 | 0 | [0.90, -0.80] | 0.00 | [0.90, -0.80] | 0.00
.....
1 | 0 | 0 | 1 | -1 | [0.90, -0.80] | 0.00 | [0.80, -0.80] | -0.10
.....
1 | 1 | 1 | 0 | 1 | [0.80, -0.80] | -0.10 | [0.90, -0.70] | 0.00
.....
=====

```

Epoch : 5, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.90, -0.70] | 0.00 | [0.90, -0.70] | -0.10
.....
0 | 1 | 0 | 0 | 0 | [0.90, -0.70] | -0.10 | [0.90, -0.70] | -0.10
.....
1 | 0 | 0 | 1 | -1 | [0.90, -0.70] | -0.10 | [0.80, -0.70] | -0.20
.....
1 | 1 | 1 | 0 | 1 | [0.80, -0.70] | -0.20 | [0.90, -0.60] | -0.10
.....
=====

```

Epoch : 6, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [0.90, -0.60] | -0.10 | [0.90, -0.60] | -0.10
.....
0 | 1 | 0 | 0 | 0 | [0.90, -0.60] | -0.10 | [0.90, -0.60] | -0.10
.....
1 | 0 | 0 | 1 | -1 | [0.90, -0.60] | -0.10 | [0.80, -0.60] | -0.20
.....
1 | 1 | 1 | 0 | 1 | [0.80, -0.60] | -0.20 | [0.90, -0.50] | -0.10
.....
=====

```

Epoch : 7, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [0.90, -0.50] | -0.10 | [0.90, -0.50] | -0.10
.....
0 | 1 | 0 | 0 | 0 | [0.90, -0.50] | -0.10 | [0.90, -0.50] | -0.10
.....
1 | 0 | 0 | 1 | -1 | [0.90, -0.50] | -0.10 | [0.80, -0.50] | -0.20
.....
1 | 1 | 1 | 1 | 0 | [0.80, -0.50] | -0.20 | [0.80, -0.50] | -0.20
.....
=====

```

Epoch : 8, learning_rate : 0.1

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.80, -0.50]	-0.20	[0.80, -0.50]	-0.20
0	1	0	0	0	[0.80, -0.50]	-0.20	[0.80, -0.50]	-0.20
1	0	0	1	-1	[0.80, -0.50]	-0.20	[0.70, -0.50]	-0.30
1	1	1	0	1	[0.70, -0.50]	-0.30	[0.80, -0.40]	-0.20

Epoch : 9, learning_rate : 0.1

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.80, -0.40]	-0.20	[0.80, -0.40]	-0.20
0	1	0	0	0	[0.80, -0.40]	-0.20	[0.80, -0.40]	-0.20
1	0	0	1	-1	[0.80, -0.40]	-0.20	[0.70, -0.40]	-0.30
1	1	1	0	1	[0.70, -0.40]	-0.30	[0.80, -0.30]	-0.20

Epoch : 10, learning_rate : 0.1

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.80, -0.30]	-0.20	[0.80, -0.30]	-0.20
0	1	0	0	0	[0.80, -0.30]	-0.20	[0.80, -0.30]	-0.20
1	0	0	1	-1	[0.80, -0.30]	-0.20	[0.70, -0.30]	-0.30
1	1	1	1	0	[0.70, -0.30]	-0.30	[0.70, -0.30]	-0.30

Epoch : 11, learning_rate : 0.1

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, -0.30]	-0.30	[0.70, -0.30]	-0.30
0	1	0	0	0	[0.70, -0.30]	-0.30	[0.70, -0.30]	-0.30
1	0	0	1	-1	[0.70, -0.30]	-0.30	[0.60, -0.30]	-0.40
1	1	1	0	1	[0.60, -0.30]	-0.40	[0.70, -0.20]	-0.30

=====
Epoch : 12, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, -0.20]	-0.30	[0.70, -0.20]	-0.30
0	1	0	0	0	[0.70, -0.20]	-0.30	[0.70, -0.20]	-0.30
1	0	0	1	-1	[0.70, -0.20]	-0.30	[0.60, -0.20]	-0.40
1	1	1	0	1	[0.60, -0.20]	-0.40	[0.70, -0.10]	-0.30

.....

=====
Epoch : 13, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, -0.10]	-0.30	[0.70, -0.10]	-0.30
0	1	0	0	0	[0.70, -0.10]	-0.30	[0.70, -0.10]	-0.30
1	0	0	1	-1	[0.70, -0.10]	-0.30	[0.60, -0.10]	-0.40
1	1	1	1	0	[0.60, -0.10]	-0.40	[0.60, -0.10]	-0.40

.....

=====
Epoch : 14, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.60, -0.10]	-0.40	[0.60, -0.10]	-0.40
0	1	0	0	0	[0.60, -0.10]	-0.40	[0.60, -0.10]	-0.40
1	0	0	1	-1	[0.60, -0.10]	-0.40	[0.50, -0.10]	-0.50
1	1	1	0	1	[0.50, -0.10]	-0.50	[0.60, -0.00]	-0.40

.....

=====
Epoch : 15, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.60, -0.00]	-0.40	[0.60, -0.00]	-0.40
0	1	0	0	0	[0.60, -0.00]	-0.40	[0.60, -0.00]	-0.40
1	0	0	1	-1	[0.60, -0.00]	-0.40	[0.50, -0.00]	-0.50

.....

1		1		1		1		0		[0.50, -0.00]		-0.50		[0.50, -0.00]		-0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	-------	--	---------------	--	-------

Epoch : 16, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[0.50, -0.00]		-0.50		[0.50, -0.00]		-0.50
0		1		0		0		0		[0.50, -0.00]		-0.50		[0.50, -0.00]		-0.50
1		0		0		1		-1		[0.50, -0.00]		-0.50		[0.40, -0.00]		-0.60
1		1		1		0		1		[0.40, -0.00]		-0.60		[0.50, 0.10]		-0.50

Epoch : 17, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[0.50, 0.10]		-0.50		[0.50, 0.10]		-0.50
0		1		0		0		0		[0.50, 0.10]		-0.50		[0.50, 0.10]		-0.50
1		0		0		1		-1		[0.50, 0.10]		-0.50		[0.40, 0.10]		-0.60
1		1		1		0		1		[0.40, 0.10]		-0.60		[0.50, 0.20]		-0.50

Epoch : 18, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[0.50, 0.20]		-0.50		[0.50, 0.20]		-0.50
0		1		0		0		0		[0.50, 0.20]		-0.50		[0.50, 0.20]		-0.50
1		0		0		1		-1		[0.50, 0.20]		-0.50		[0.40, 0.20]		-0.60
1		1		1		1		0		[0.40, 0.20]		-0.60		[0.40, 0.20]		-0.60

Epoch : 19, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[0.40, 0.20]		-0.60		[0.40, 0.20]		-0.60
0		1		0		0		0		[0.40, 0.20]		-0.60		[0.40, 0.20]		-0.60

1		0		0		0		0		[0.40, 0.20]		-0.60		[0.40, 0.20]		-0.60
1		1		1		1		0		[0.40, 0.20]		-0.60		[0.40, 0.20]		-0.60

Epoch : 1, learning_rate : 0.2

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.30
0		1		0		0		0		[1.00, -1.00]		0.30		[1.00, -1.00]		0.30
1		0		0		1		-1		[1.00, -1.00]		0.30		[0.80, -1.00]		0.10
1		1		1		0		1		[0.80, -1.00]		0.10		[1.00, -0.80]		0.30

Epoch : 2, learning_rate : 0.2

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -0.80]		0.30		[1.00, -0.80]		0.10
0		1		0		0		0		[1.00, -0.80]		0.10		[1.00, -0.80]		0.10
1		0		0		1		-1		[1.00, -0.80]		0.10		[0.80, -0.80]		-0.10
1		1		1		0		1		[0.80, -0.80]		-0.10		[1.00, -0.60]		0.10

Epoch : 3, learning_rate : 0.2

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -0.60]		0.10		[1.00, -0.60]		-0.10
0		1		0		0		0		[1.00, -0.60]		-0.10		[1.00, -0.60]		-0.10
1		0		0		1		-1		[1.00, -0.60]		-0.10		[0.80, -0.60]		-0.30
1		1		1		0		1		[0.80, -0.60]		-0.30		[1.00, -0.40]		-0.10

Epoch : 4, learning_rate : 0.2

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[1.00, -0.40]		-0.10		[1.00, -0.40]		-0.10

0	1	0	0	0	[1.00, -0.40]	-0.10	[1.00, -0.40]	-0.10
1	0	0	1	-1	[1.00, -0.40]	-0.10	[0.80, -0.40]	-0.30
1	1	1	1	0	[0.80, -0.40]	-0.30	[0.80, -0.40]	-0.30

Epoch : 5, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.80, -0.40]	-0.30	[0.80, -0.40]	-0.30
0	1	0	0	0	[0.80, -0.40]	-0.30	[0.80, -0.40]	-0.30
1	0	0	1	-1	[0.80, -0.40]	-0.30	[0.60, -0.40]	-0.50
1	1	1	0	1	[0.60, -0.40]	-0.50	[0.80, -0.20]	-0.30

Epoch : 6, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.80, -0.20]	-0.30	[0.80, -0.20]	-0.30
0	1	0	0	0	[0.80, -0.20]	-0.30	[0.80, -0.20]	-0.30
1	0	0	1	-1	[0.80, -0.20]	-0.30	[0.60, -0.20]	-0.50
1	1	1	0	1	[0.60, -0.20]	-0.50	[0.80, -0.00]	-0.30

Epoch : 7, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.80, -0.00]	-0.30	[0.80, -0.00]	-0.30
0	1	0	0	0	[0.80, -0.00]	-0.30	[0.80, -0.00]	-0.30
1	0	0	1	-1	[0.80, -0.00]	-0.30	[0.60, -0.00]	-0.50
1	1	1	1	0	[0.60, -0.00]	-0.50	[0.60, -0.00]	-0.50

Epoch : 8, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	0	0	[0.60, -0.00]	-0.50	[0.60, -0.00]	-0.50
.....								
0	1	0	0	0	[0.60, -0.00]	-0.50	[0.60, -0.00]	-0.50
.....								
1	0	0	1	-1	[0.60, -0.00]	-0.50	[0.40, -0.00]	-0.70
.....								
1	1	1	0	1	[0.40, -0.00]	-0.70	[0.60, 0.20]	-0.50
.....								

=====
Epoch : 9, learning_rate : 0.2
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	0	0	[0.60, 0.20]	-0.50	[0.60, 0.20]	-0.50
.....								
0	1	0	0	0	[0.60, 0.20]	-0.50	[0.60, 0.20]	-0.50
.....								
1	0	0	1	-1	[0.60, 0.20]	-0.50	[0.40, 0.20]	-0.70
.....								
1	1	1	0	1	[0.40, 0.20]	-0.70	[0.60, 0.40]	-0.50
.....								

=====
Epoch : 10, learning_rate : 0.2
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	0	0	[0.60, 0.40]	-0.50	[0.60, 0.40]	-0.50
.....								
0	1	0	0	0	[0.60, 0.40]	-0.50	[0.60, 0.40]	-0.50
.....								
1	0	0	1	-1	[0.60, 0.40]	-0.50	[0.40, 0.40]	-0.70
.....								
1	1	1	1	0	[0.40, 0.40]	-0.70	[0.40, 0.40]	-0.70
.....								

=====
Epoch : 11, learning_rate : 0.2
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	0	0	[0.40, 0.40]	-0.70	[0.40, 0.40]	-0.70
.....								
0	1	0	0	0	[0.40, 0.40]	-0.70	[0.40, 0.40]	-0.70
.....								
1	0	0	0	0	[0.40, 0.40]	-0.70	[0.40, 0.40]	-0.70
.....								
1	1	1	1	0	[0.40, 0.40]	-0.70	[0.40, 0.40]	-0.70
.....								

=====
Epoch : 1, learning_rate : 0.3
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.20
0	1	0	0	0	[1.00, -1.00]	0.20	[1.00, -1.00]	0.20
1	0	0	1	-1	[1.00, -1.00]	0.20	[0.70, -1.00]	-0.10
1	1	1	0	1	[0.70, -1.00]	-0.10	[1.00, -0.70]	0.20

Epoch : 2, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.70]	0.20	[1.00, -0.70]	-0.10
0	1	0	0	0	[1.00, -0.70]	-0.10	[1.00, -0.70]	-0.10
1	0	0	1	-1	[1.00, -0.70]	-0.10	[0.70, -0.70]	-0.40
1	1	1	0	1	[0.70, -0.70]	-0.40	[1.00, -0.40]	-0.10

Epoch : 3, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, -0.40]	-0.10	[1.00, -0.40]	-0.10
0	1	0	0	0	[1.00, -0.40]	-0.10	[1.00, -0.40]	-0.10
1	0	0	1	-1	[1.00, -0.40]	-0.10	[0.70, -0.40]	-0.40
1	1	1	0	1	[0.70, -0.40]	-0.40	[1.00, -0.10]	-0.10

Epoch : 4, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, -0.10]	-0.10	[1.00, -0.10]	-0.10
0	1	0	0	0	[1.00, -0.10]	-0.10	[1.00, -0.10]	-0.10
1	0	0	1	-1	[1.00, -0.10]	-0.10	[0.70, -0.10]	-0.40
1	1	1	1	0	[0.70, -0.10]	-0.40	[0.70, -0.10]	-0.40

Epoch : 5, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, -0.10]	-0.40	[0.70, -0.10]	-0.40
0	1	0	0	0	[0.70, -0.10]	-0.40	[0.70, -0.10]	-0.40
1	0	0	1	-1	[0.70, -0.10]	-0.40	[0.40, -0.10]	-0.70
1	1	1	0	1	[0.40, -0.10]	-0.70	[0.70, 0.20]	-0.40

Epoch : 6, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, 0.20]	-0.40	[0.70, 0.20]	-0.40
0	1	0	0	0	[0.70, 0.20]	-0.40	[0.70, 0.20]	-0.40
1	0	0	1	-1	[0.70, 0.20]	-0.40	[0.40, 0.20]	-0.70
1	1	1	0	1	[0.40, 0.20]	-0.70	[0.70, 0.50]	-0.40

Epoch : 7, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, 0.50]	-0.40	[0.70, 0.50]	-0.40
0	1	0	1	-1	[0.70, 0.50]	-0.40	[0.70, 0.20]	-0.70
1	0	0	1	-1	[0.70, 0.20]	-0.70	[0.40, 0.20]	-1.00
1	1	1	0	1	[0.40, 0.20]	-1.00	[0.70, 0.50]	-0.70

Epoch : 8, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, 0.50]	-0.70	[0.70, 0.50]	-0.70
0	1	0	0	0	[0.70, 0.50]	-0.70	[0.70, 0.50]	-0.70
1	0	0	1	-1	[0.70, 0.50]	-0.70	[0.40, 0.50]	-1.00
1	1	1	0	1	[0.40, 0.50]	-1.00	[0.70, 0.80]	-0.70

.....
=====

Epoch : 9, learning_rate : 0.3

=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, 0.80]	-0.70	[0.70, 0.80]	-0.70
0	1	0	1	-1	[0.70, 0.80]	-0.70	[0.70, 0.50]	-1.00
1	0	0	0	0	[0.70, 0.50]	-1.00	[0.70, 0.50]	-1.00
1	1	1	1	0	[0.70, 0.50]	-1.00	[0.70, 0.50]	-1.00

.....

=====

Epoch : 10, learning_rate : 0.3

=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.70, 0.50]	-1.00	[0.70, 0.50]	-1.00
0	1	0	0	0	[0.70, 0.50]	-1.00	[0.70, 0.50]	-1.00
1	0	0	0	0	[0.70, 0.50]	-1.00	[0.70, 0.50]	-1.00
1	1	1	1	0	[0.70, 0.50]	-1.00	[0.70, 0.50]	-1.00

.....

=====

Epoch : 1, learning_rate : 0.4

=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.10
0	1	0	0	0	[1.00, -1.00]	0.10	[1.00, -1.00]	0.10
1	0	0	1	-1	[1.00, -1.00]	0.10	[0.60, -1.00]	-0.30
1	1	1	0	1	[0.60, -1.00]	-0.30	[1.00, -0.60]	0.10

.....

=====

Epoch : 2, learning_rate : 0.4

=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.60]	0.10	[1.00, -0.60]	-0.30
0	1	0	0	0	[1.00, -0.60]	-0.30	[1.00, -0.60]	-0.30
1	0	0	1	-1	[1.00, -0.60]	-0.30	[0.60, -0.60]	-0.70

```
.....
1 | 1 | 1 | 0 | 1 | [0.60, -0.60] | -0.70 | [1.00, -0.20] | -0.30
.....
```

=====
Epoch : 3, learning_rate : 0.4
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, -0.20] | -0.30 | [1.00, -0.20] | -0.30
.....
0 | 1 | 0 | 0 | 0 | [1.00, -0.20] | -0.30 | [1.00, -0.20] | -0.30
.....
1 | 0 | 0 | 1 | -1 | [1.00, -0.20] | -0.30 | [0.60, -0.20] | -0.70
.....
1 | 1 | 1 | 0 | 1 | [0.60, -0.20] | -0.70 | [1.00, 0.20] | -0.30
.....
```

=====
Epoch : 4, learning_rate : 0.4
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.20] | -0.30 | [1.00, 0.20] | -0.30
.....
0 | 1 | 0 | 0 | 0 | [1.00, 0.20] | -0.30 | [1.00, 0.20] | -0.30
.....
1 | 0 | 0 | 1 | -1 | [1.00, 0.20] | -0.30 | [0.60, 0.20] | -0.70
.....
1 | 1 | 1 | 1 | 0 | [0.60, 0.20] | -0.70 | [0.60, 0.20] | -0.70
.....
```

=====
Epoch : 5, learning_rate : 0.4
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [0.60, 0.20] | -0.70 | [0.60, 0.20] | -0.70
.....
0 | 1 | 0 | 0 | 0 | [0.60, 0.20] | -0.70 | [0.60, 0.20] | -0.70
.....
1 | 0 | 0 | 0 | 0 | [0.60, 0.20] | -0.70 | [0.60, 0.20] | -0.70
.....
1 | 1 | 1 | 1 | 0 | [0.60, 0.20] | -0.70 | [0.60, 0.20] | -0.70
.....
```

=====
Epoch : 1, learning_rate : 0.5
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.00
.....
0 | 1 | 0 | 0 | 0 | [1.00, -1.00] | 0.00 | [1.00, -1.00] | 0.00
.....
```

```

.....
1 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.00 | [0.50, -1.00] | -0.50
.....
1 | 1 | 1 | 0 | 1 | [0.50, -1.00] | -0.50 | [1.00, -0.50] | 0.00
.....
=====

```

Epoch : 2, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.50] | 0.00 | [1.00, -0.50] | -0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, -0.50] | -0.50 | [1.00, -0.50] | -0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, -0.50] | -0.50 | [0.50, -0.50] | -1.00
.....
1 | 1 | 1 | 0 | 1 | [0.50, -0.50] | -1.00 | [1.00, 0.00] | -0.50
.....
=====

```

Epoch : 3, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.00] | -0.50 | [1.00, 0.00] | -0.50
.....
0 | 1 | 0 | 0 | 0 | [1.00, 0.00] | -0.50 | [1.00, 0.00] | -0.50
.....
1 | 0 | 0 | 1 | -1 | [1.00, 0.00] | -0.50 | [0.50, 0.00] | -1.00
.....
1 | 1 | 1 | 0 | 1 | [0.50, 0.00] | -1.00 | [1.00, 0.50] | -0.50
.....
=====

```

Epoch : 4, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....
0 | 1 | 0 | 1 | -1 | [1.00, 0.50] | -0.50 | [1.00, 0.00] | -1.00
.....
1 | 0 | 0 | 1 | -1 | [1.00, 0.00] | -1.00 | [0.50, 0.00] | -1.50
.....
1 | 1 | 1 | 0 | 1 | [0.50, 0.00] | -1.50 | [1.00, 0.50] | -1.00
.....
=====

```

Epoch : 5, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.50] | -1.00 | [1.00, 0.50] | -1.00

```

```

.....
0 | 1 | 0 | 0 | 0 | [1.00, 0.50] | -1.00 | [1.00, 0.50] | -1.00
.....
1 | 0 | 0 | 1 | -1 | [1.00, 0.50] | -1.00 | [0.50, 0.50] | -1.50
.....
1 | 1 | 1 | 0 | 1 | [0.50, 0.50] | -1.50 | [1.00, 1.00] | -1.00
.....

```

=====
Epoch : 6, learning_rate : 0.5
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 1.00]	-1.00	[1.00, 1.00]	-1.00
0	1	0	1	-1	[1.00, 1.00]	-1.00	[1.00, 0.50]	-1.50
1	0	0	0	0	[1.00, 0.50]	-1.50	[1.00, 0.50]	-1.50
1	1	1	1	0	[1.00, 0.50]	-1.50	[1.00, 0.50]	-1.50

=====
Epoch : 7, learning_rate : 0.5
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.50]	-1.50	[1.00, 0.50]	-1.50
0	1	0	0	0	[1.00, 0.50]	-1.50	[1.00, 0.50]	-1.50
1	0	0	0	0	[1.00, 0.50]	-1.50	[1.00, 0.50]	-1.50
1	1	1	1	0	[1.00, 0.50]	-1.50	[1.00, 0.50]	-1.50

=====
Epoch : 1, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.10
0	1	0	0	0	[1.00, -1.00]	-0.10	[1.00, -1.00]	-0.10
1	0	0	1	-1	[1.00, -1.00]	-0.10	[0.40, -1.00]	-0.70
1	1	1	0	1	[0.40, -1.00]	-0.70	[1.00, -0.40]	-0.10

=====
Epoch : 2, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 0 | 0 | [1.00, -0.40] | -0.10 | [1.00, -0.40] | -0.10
.....
0 | 1 | 0 | 0 | 0 | [1.00, -0.40] | -0.10 | [1.00, -0.40] | -0.10
.....
1 | 0 | 0 | 1 | -1 | [1.00, -0.40] | -0.10 | [0.40, -0.40] | -0.70
.....
1 | 1 | 1 | 0 | 1 | [0.40, -0.40] | -0.70 | [1.00, 0.20] | -0.10
.....
=====

```

Epoch : 3, learning_rate : 0.6

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....
0 | 1 | 0 | 1 | -1 | [1.00, 0.20] | -0.10 | [1.00, -0.40] | -0.70
.....
1 | 0 | 0 | 1 | -1 | [1.00, -0.40] | -0.70 | [0.40, -0.40] | -1.30
.....
1 | 1 | 1 | 0 | 1 | [0.40, -0.40] | -1.30 | [1.00, 0.20] | -0.70
.....
=====

```

Epoch : 4, learning_rate : 0.6

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.20] | -0.70 | [1.00, 0.20] | -0.70
.....
0 | 1 | 0 | 0 | 0 | [1.00, 0.20] | -0.70 | [1.00, 0.20] | -0.70
.....
1 | 0 | 0 | 1 | -1 | [1.00, 0.20] | -0.70 | [0.40, 0.20] | -1.30
.....
1 | 1 | 1 | 0 | 1 | [0.40, 0.20] | -1.30 | [1.00, 0.80] | -0.70
.....
=====

```

Epoch : 5, learning_rate : 0.6

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.80] | -0.70 | [1.00, 0.80] | -0.70
.....
0 | 1 | 0 | 1 | -1 | [1.00, 0.80] | -0.70 | [1.00, 0.20] | -1.30
.....
1 | 0 | 0 | 0 | 0 | [1.00, 0.20] | -1.30 | [1.00, 0.20] | -1.30
.....
1 | 1 | 1 | 0 | 1 | [1.00, 0.20] | -1.30 | [1.60, 0.80] | -0.70
.....
=====

```

Epoch : 6, learning_rate : 0.6

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.60, 0.80]	-0.70	[1.60, 0.80]	-0.70
0	1	0	1	-1	[1.60, 0.80]	-0.70	[1.60, 0.20]	-1.30
1	0	0	1	-1	[1.60, 0.20]	-1.30	[1.00, 0.20]	-1.90
1	1	1	0	1	[1.00, 0.20]	-1.90	[1.60, 0.80]	-1.30

Epoch : 7, learning_rate : 0.6

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.60, 0.80]	-1.30	[1.60, 0.80]	-1.30
0	1	0	0	0	[1.60, 0.80]	-1.30	[1.60, 0.80]	-1.30
1	0	0	1	-1	[1.60, 0.80]	-1.30	[1.00, 0.80]	-1.90
1	1	1	0	1	[1.00, 0.80]	-1.90	[1.60, 1.40]	-1.30

Epoch : 8, learning_rate : 0.6

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.60, 1.40]	-1.30	[1.60, 1.40]	-1.30
0	1	0	1	-1	[1.60, 1.40]	-1.30	[1.60, 0.80]	-1.90
1	0	0	0	0	[1.60, 0.80]	-1.90	[1.60, 0.80]	-1.90
1	1	1	1	0	[1.60, 0.80]	-1.90	[1.60, 0.80]	-1.90

Epoch : 9, learning_rate : 0.6

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.60, 0.80]	-1.90	[1.60, 0.80]	-1.90
0	1	0	0	0	[1.60, 0.80]	-1.90	[1.60, 0.80]	-1.90
1	0	0	0	0	[1.60, 0.80]	-1.90	[1.60, 0.80]	-1.90
1	1	1	1	0	[1.60, 0.80]	-1.90	[1.60, 0.80]	-1.90

=====
Epoch : 1, learning_rate : 0.7
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.20
0	1	0	0	0	[1.00, -1.00]	-0.20	[1.00, -1.00]	-0.20
1	0	0	1	-1	[1.00, -1.00]	-0.20	[0.30, -1.00]	-0.90
1	1	1	0	1	[0.30, -1.00]	-0.90	[1.00, -0.30]	-0.20

.....

=====
Epoch : 2, learning_rate : 0.7
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, -0.30]	-0.20	[1.00, -0.30]	-0.20
0	1	0	0	0	[1.00, -0.30]	-0.20	[1.00, -0.30]	-0.20
1	0	0	1	-1	[1.00, -0.30]	-0.20	[0.30, -0.30]	-0.90
1	1	1	0	1	[0.30, -0.30]	-0.90	[1.00, 0.40]	-0.20

.....

=====
Epoch : 3, learning_rate : 0.7
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.40]	-0.20	[1.00, 0.40]	-0.20
0	1	0	1	-1	[1.00, 0.40]	-0.20	[1.00, -0.30]	-0.90
1	0	0	1	-1	[1.00, -0.30]	-0.90	[0.30, -0.30]	-1.60
1	1	1	0	1	[0.30, -0.30]	-1.60	[1.00, 0.40]	-0.90

.....

=====
Epoch : 4, learning_rate : 0.7
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.40]	-0.90	[1.00, 0.40]	-0.90
0	1	0	0	0	[1.00, 0.40]	-0.90	[1.00, 0.40]	-0.90
1	0	0	1	-1	[1.00, 0.40]	-0.90	[0.30, 0.40]	-1.60

.....

1		1		1		0		1		[0.30, 0.40]		-1.60		[1.00, 1.10]		-0.90
---	--	---	--	---	--	---	--	---	--	--------------	--	-------	--	--------------	--	-------

Epoch : 5, learning_rate : 0.7

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[1.00, 1.10]		-0.90		[1.00, 1.10]		-0.90
0		1		0		1		-1		[1.00, 1.10]		-0.90		[1.00, 0.40]		-1.60
1		0		0		0		0		[1.00, 0.40]		-1.60		[1.00, 0.40]		-1.60
1		1		1		0		1		[1.00, 0.40]		-1.60		[1.70, 1.10]		-0.90

Epoch : 6, learning_rate : 0.7

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[1.70, 1.10]		-0.90		[1.70, 1.10]		-0.90
0		1		0		1		-1		[1.70, 1.10]		-0.90		[1.70, 0.40]		-1.60
1		0		0		1		-1		[1.70, 0.40]		-1.60		[1.00, 0.40]		-2.30
1		1		1		0		1		[1.00, 0.40]		-2.30		[1.70, 1.10]		-1.60

Epoch : 7, learning_rate : 0.7

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[1.70, 1.10]		-1.60		[1.70, 1.10]		-1.60
0		1		0		0		0		[1.70, 1.10]		-1.60		[1.70, 1.10]		-1.60
1		0		0		1		-1		[1.70, 1.10]		-1.60		[1.00, 1.10]		-2.30
1		1		1		0		1		[1.00, 1.10]		-2.30		[1.70, 1.80]		-1.60

Epoch : 8, learning_rate : 0.7

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[1.70, 1.80]		-1.60		[1.70, 1.80]		-1.60
0		1		0		1		-1		[1.70, 1.80]		-1.60		[1.70, 1.10]		-2.30

1		0		0		0		0		[1.70, 1.10]		-2.30		[1.70, 1.10]		-2.30
.....																
1		1		1		1		0		[1.70, 1.10]		-2.30		[1.70, 1.10]		-2.30
.....																

=====
Epoch : 9, learning_rate : 0.7

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new

0		0		0		0		0		[1.70, 1.10]		-2.30		[1.70, 1.10]		-2.30
.....																
0		1		0		0		0		[1.70, 1.10]		-2.30		[1.70, 1.10]		-2.30
.....																
1		0		0		0		0		[1.70, 1.10]		-2.30		[1.70, 1.10]		-2.30
.....																
1		1		1		1		0		[1.70, 1.10]		-2.30		[1.70, 1.10]		-2.30
.....																

=====
Epoch : 1, learning_rate : 0.8

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new

0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		-0.30
.....																
0		1		0		0		0		[1.00, -1.00]		-0.30		[1.00, -1.00]		-0.30
.....																
1		0		0		1		-1		[1.00, -1.00]		-0.30		[0.20, -1.00]		-1.10
.....																
1		1		1		0		1		[0.20, -1.00]		-1.10		[1.00, -0.20]		-0.30
.....																

=====
Epoch : 2, learning_rate : 0.8

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new

0		0		0		0		0		[1.00, -0.20]		-0.30		[1.00, -0.20]		-0.30
.....																
0		1		0		0		0		[1.00, -0.20]		-0.30		[1.00, -0.20]		-0.30
.....																
1		0		0		1		-1		[1.00, -0.20]		-0.30		[0.20, -0.20]		-1.10
.....																
1		1		1		0		1		[0.20, -0.20]		-1.10		[1.00, 0.60]		-0.30
.....																

=====
Epoch : 3, learning_rate : 0.8

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new

0		0		0		0		0		[1.00, 0.60]		-0.30		[1.00, 0.60]		-0.30
.....																

0	1	0	1	-1	[1.00, 0.60]	-0.30	[1.00, -0.20]	-1.10
1	0	0	0	0	[1.00, -0.20]	-1.10	[1.00, -0.20]	-1.10
1	1	1	0	1	[1.00, -0.20]	-1.10	[1.80, 0.60]	-0.30

Epoch : 4, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.80, 0.60]	-0.30	[1.80, 0.60]	-0.30
0	1	0	1	-1	[1.80, 0.60]	-0.30	[1.80, -0.20]	-1.10
1	0	0	1	-1	[1.80, -0.20]	-1.10	[1.00, -0.20]	-1.90
1	1	1	0	1	[1.00, -0.20]	-1.90	[1.80, 0.60]	-1.10

Epoch : 5, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.80, 0.60]	-1.10	[1.80, 0.60]	-1.10
0	1	0	0	0	[1.80, 0.60]	-1.10	[1.80, 0.60]	-1.10
1	0	0	1	-1	[1.80, 0.60]	-1.10	[1.00, 0.60]	-1.90
1	1	1	0	1	[1.00, 0.60]	-1.90	[1.80, 1.40]	-1.10

Epoch : 6, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.80, 1.40]	-1.10	[1.80, 1.40]	-1.10
0	1	0	1	-1	[1.80, 1.40]	-1.10	[1.80, 0.60]	-1.90
1	0	0	0	0	[1.80, 0.60]	-1.90	[1.80, 0.60]	-1.90
1	1	1	1	0	[1.80, 0.60]	-1.90	[1.80, 0.60]	-1.90

Epoch : 7, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	0	0	[1.80, 0.60]	-1.90	[1.80, 0.60]	-1.90
0	1	0	0	0	[1.80, 0.60]	-1.90	[1.80, 0.60]	-1.90
1	0	0	0	0	[1.80, 0.60]	-1.90	[1.80, 0.60]	-1.90
1	1	1	1	0	[1.80, 0.60]	-1.90	[1.80, 0.60]	-1.90

Epoch : 1, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.40
0	1	0	0	0	[1.00, -1.00]	-0.40	[1.00, -1.00]	-0.40
1	0	0	1	-1	[1.00, -1.00]	-0.40	[0.10, -1.00]	-1.30
1	1	1	0	1	[0.10, -1.00]	-1.30	[1.00, -0.10]	-0.40

Epoch : 2, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, -0.10]	-0.40	[1.00, -0.10]	-0.40
0	1	0	0	0	[1.00, -0.10]	-0.40	[1.00, -0.10]	-0.40
1	0	0	1	-1	[1.00, -0.10]	-0.40	[0.10, -0.10]	-1.30
1	1	1	0	1	[0.10, -0.10]	-1.30	[1.00, 0.80]	-0.40

Epoch : 3, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.80]	-0.40	[1.00, 0.80]	-0.40
0	1	0	1	-1	[1.00, 0.80]	-0.40	[1.00, -0.10]	-1.30
1	0	0	0	0	[1.00, -0.10]	-1.30	[1.00, -0.10]	-1.30
1	1	1	0	1	[1.00, -0.10]	-1.30	[1.90, 0.80]	-0.40

Epoch : 4, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.90, 0.80]	-0.40	[1.90, 0.80]	-0.40
0	1	0	1	-1	[1.90, 0.80]	-0.40	[1.90, -0.10]	-1.30
1	0	0	1	-1	[1.90, -0.10]	-1.30	[1.00, -0.10]	-2.20
1	1	1	0	1	[1.00, -0.10]	-2.20	[1.90, 0.80]	-1.30

Epoch : 5, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.90, 0.80]	-1.30	[1.90, 0.80]	-1.30
0	1	0	0	0	[1.90, 0.80]	-1.30	[1.90, 0.80]	-1.30
1	0	0	1	-1	[1.90, 0.80]	-1.30	[1.00, 0.80]	-2.20
1	1	1	0	1	[1.00, 0.80]	-2.20	[1.90, 1.70]	-1.30

Epoch : 6, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.90, 1.70]	-1.30	[1.90, 1.70]	-1.30
0	1	0	1	-1	[1.90, 1.70]	-1.30	[1.90, 0.80]	-2.20
1	0	0	0	0	[1.90, 0.80]	-2.20	[1.90, 0.80]	-2.20
1	1	1	1	0	[1.90, 0.80]	-2.20	[1.90, 0.80]	-2.20

Epoch : 7, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.90, 0.80]	-2.20	[1.90, 0.80]	-2.20
0	1	0	0	0	[1.90, 0.80]	-2.20	[1.90, 0.80]	-2.20
1	0	0	0	0	[1.90, 0.80]	-2.20	[1.90, 0.80]	-2.20
1	1	1	1	0	[1.90, 0.80]	-2.20	[1.90, 0.80]	-2.20

Epoch : 1, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.50
0	1	0	0	0	[1.00, -1.00]	-0.50	[1.00, -1.00]	-0.50
1	0	0	1	-1	[1.00, -1.00]	-0.50	[0.00, -1.00]	-1.50
1	1	1	0	1	[0.00, -1.00]	-1.50	[1.00, 0.00]	-0.50

Epoch : 2, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.00]	-0.50	[1.00, 0.00]	-0.50
0	1	0	0	0	[1.00, 0.00]	-0.50	[1.00, 0.00]	-0.50
1	0	0	1	-1	[1.00, 0.00]	-0.50	[0.00, 0.00]	-1.50
1	1	1	0	1	[0.00, 0.00]	-1.50	[1.00, 1.00]	-0.50

Epoch : 3, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 1.00]	-0.50	[1.00, 1.00]	-0.50
0	1	0	1	-1	[1.00, 1.00]	-0.50	[1.00, 0.00]	-1.50
1	0	0	0	0	[1.00, 0.00]	-1.50	[1.00, 0.00]	-1.50
1	1	1	0	1	[1.00, 0.00]	-1.50	[2.00, 1.00]	-0.50

Epoch : 4, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[2.00, 1.00]	-0.50	[2.00, 1.00]	-0.50
0	1	0	1	-1	[2.00, 1.00]	-0.50	[2.00, 0.00]	-1.50
1	0	0	1	-1	[2.00, 0.00]	-1.50	[1.00, 0.00]	-2.50
1	1	1	0	1	[1.00, 0.00]	-2.50	[2.00, 1.00]	-1.50

```

.....
=====
Epoch : 5, learning_rate : 1.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0     | 0   | [2.00, 1.00]   | -1.50 | [2.00, 1.00]   | -1.50
.....
0 | 1 | 0 | 0     | 0   | [2.00, 1.00]   | -1.50 | [2.00, 1.00]   | -1.50
.....
1 | 0 | 0 | 1     | -1  | [2.00, 1.00]   | -1.50 | [1.00, 1.00]   | -2.50
.....
1 | 1 | 1 | 0     | 1   | [1.00, 1.00]   | -2.50 | [2.00, 2.00]   | -1.50
.....
=====

```

```

Epoch : 6, learning_rate : 1.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0     | 0   | [2.00, 2.00]   | -1.50 | [2.00, 2.00]   | -1.50
.....
0 | 1 | 0 | 1     | -1  | [2.00, 2.00]   | -1.50 | [2.00, 1.00]   | -2.50
.....
1 | 0 | 0 | 0     | 0   | [2.00, 1.00]   | -2.50 | [2.00, 1.00]   | -2.50
.....
1 | 1 | 1 | 1     | 0   | [2.00, 1.00]   | -2.50 | [2.00, 1.00]   | -2.50
.....
=====

```

```

Epoch : 7, learning_rate : 1.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0     | 0   | [2.00, 1.00]   | -2.50 | [2.00, 1.00]   | -2.50
.....
0 | 1 | 0 | 0     | 0   | [2.00, 1.00]   | -2.50 | [2.00, 1.00]   | -2.50
.....
1 | 0 | 0 | 0     | 0   | [2.00, 1.00]   | -2.50 | [2.00, 1.00]   | -2.50
.....
1 | 1 | 1 | 1     | 0   | [2.00, 1.00]   | -2.50 | [2.00, 1.00]   | -2.50
.....
=====

```

AND Gate summery

```

=====
| LR | Converged | Epochs | Updates | Weights          | Bias   | Margin |
=====
| 0.0 | False    | 200    | 400     | [1.00, -1.00]   | 0.50  | 0.35  |
| 0.1 | True     | 19     | 35      | [0.40, 0.20]   | -0.60 | 0.00  |
| 0.2 | True     | 11     | 20      | [0.40, 0.40]   | -0.70 | 0.18  |
| 0.3 | True     | 10     | 19      | [0.70, 0.50]   | -1.00 | 0.23  |
| 0.4 | True     | 5      | 9       | [0.60, 0.20]   | -0.70 | 0.16  |
=====

```

0.5	True	7	14	[1.00, 0.50]	-1.50	0.00
0.6	True	9	18	[1.60, 0.80]	-1.90	0.17
0.7	True	9	18	[1.70, 1.10]	-2.30	0.25
0.8	True	7	13	[1.80, 0.60]	-1.90	0.05
0.9	True	7	13	[1.90, 0.80]	-2.20	0.15
1.0	True	7	13	[2.00, 1.00]	-2.50	0.22

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

X1	X2	Ac	Pred	Err	W old	b old	W new	b new
----	----	----	------	-----	-------	-------	-------	-------

0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

X1	X2	Ac	Pred	Err	W old	b old	W new	b new
----	----	----	------	-----	-------	-------	-------	-------

0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

X1	X2	Ac	Pred	Err	W old	b old	W new	b new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 5, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 6, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 7, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 8, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 9, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 10, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 11, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 12, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 13, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 14, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 15, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 16, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 17, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 18, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 19, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 20, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 21, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 22, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 23, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 24, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 25, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 26, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 27, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 28, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 29, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 30, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 31, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 32, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 33, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 34, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 35, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 36, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 37, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 38, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 39, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 40, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 41, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 42, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 43, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 44, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 45, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 46, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```

.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 47, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 48, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 49, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 50, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50

```

```

.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 51, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 52, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 53, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 54, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 55, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 56, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 57, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 58, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 59, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 60, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 61, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

=====
Epoch : 62, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 63, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 64, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 65, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 66, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 67, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 68, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 69, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 70, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 71, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 72, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 73, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 74, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 75, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 76, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 77, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 78, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 79, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 80, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 81, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 82, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 83, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 84, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 85, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 86, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 87, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 88, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 89, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 90, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 91, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

```

Epoch : 92, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 93, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 94, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 95, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 96, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```

.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 97, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 98, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 99, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 100, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50

```

```

.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 101, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 102, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 103, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 104, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 105, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 106, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 107, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 108, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 109, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 110, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

Epoch : 111, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

=====
Epoch : 112, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 113, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 114, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 115, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 116, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 117, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 118, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 119, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 120, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 121, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 122, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 123, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 124, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 125, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 126, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 127, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 128, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 129, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 130, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

Epoch : 131, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 132, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 133, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 134, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 135, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 136, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 137, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 138, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 139, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....

```

```

=====
Epoch : 140, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....

```

```

=====
Epoch : 141, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....

```

```

=====
Epoch : 142, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 143, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 144, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 145, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 146, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 147, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 148, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 149, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

=====
Epoch : 150, learning_rate : 0.0
=====

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
```

```

.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 151, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 152, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 153, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

.....

=====
Epoch : 154, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 155, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 156, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 157, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 158, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 159, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 160, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 161, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0    | 1  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1    | 0  | [1.00, -1.00]  | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

=====
Epoch : 162, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 163, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 164, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

=====
Epoch : 165, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 166, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 167, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 168, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		0		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
1		1		1		1		0		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

Epoch : 169, learning_rate : 0.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50
0		1		1		0		1		[1.00, -1.00]		0.50		[1.00, -1.00]		0.50

1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 170, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 171, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 172, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 173, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 174, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 175, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 176, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 177, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 178, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 179, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 180, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
.....								

=====
Epoch : 181, learning_rate : 0.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 182, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 183, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 184, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 185, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 186, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 187, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 188, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 189, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 190, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 191, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 1  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
=====

```

```

Epoch : 192, learning_rate : 0.0
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.50
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 193, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 194, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 195, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```
=====
Epoch : 196, learning_rate : 0.0
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
```

```

.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 197, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 198, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 199, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

Epoch : 200, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50

```

```

.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....

```

=====
Epoch : 1, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.40
0	1	1	0	1	[1.00, -1.00]	0.40	[1.00, -0.90]	0.50
1	0	1	1	0	[1.00, -0.90]	0.50	[1.00, -0.90]	0.50
1	1	1	1	0	[1.00, -0.90]	0.50	[1.00, -0.90]	0.50

.....

=====
Epoch : 2, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.90]	0.50	[1.00, -0.90]	0.40
0	1	1	0	1	[1.00, -0.90]	0.40	[1.00, -0.80]	0.50
1	0	1	1	0	[1.00, -0.80]	0.50	[1.00, -0.80]	0.50
1	1	1	1	0	[1.00, -0.80]	0.50	[1.00, -0.80]	0.50

.....

=====
Epoch : 3, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.80]	0.50	[1.00, -0.80]	0.40
0	1	1	0	1	[1.00, -0.80]	0.40	[1.00, -0.70]	0.50
1	0	1	1	0	[1.00, -0.70]	0.50	[1.00, -0.70]	0.50
1	1	1	1	0	[1.00, -0.70]	0.50	[1.00, -0.70]	0.50

.....

=====
Epoch : 4, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.70] | 0.50 | [1.00, -0.70] | 0.40
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.70] | 0.40 | [1.00, -0.60] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.60] | 0.50 | [1.00, -0.60] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -0.60] | 0.50 | [1.00, -0.60] | 0.50
.....
=====

```

Epoch : 5, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.60] | 0.50 | [1.00, -0.60] | 0.40
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.60] | 0.40 | [1.00, -0.50] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.50] | 0.50 | [1.00, -0.50] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -0.50] | 0.50 | [1.00, -0.50] | 0.50
.....
=====

```

Epoch : 6, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.50] | 0.50 | [1.00, -0.50] | 0.40
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.50] | 0.40 | [1.00, -0.40] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.40] | 0.50 | [1.00, -0.40] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -0.40] | 0.50 | [1.00, -0.40] | 0.50
.....
=====

```

Epoch : 7, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.40] | 0.50 | [1.00, -0.40] | 0.40
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.40] | 0.40 | [1.00, -0.30] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.30] | 0.50 | [1.00, -0.30] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -0.30] | 0.50 | [1.00, -0.30] | 0.50
.....
=====

```

Epoch : 8, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -0.30]  | 0.50  | [1.00, -0.30]  | 0.40
.....
0  | 1  | 1  | 1    | 0   | [1.00, -0.30]  | 0.40  | [1.00, -0.30]  | 0.40
.....
1  | 0  | 1  | 1    | 0   | [1.00, -0.30]  | 0.40  | [1.00, -0.30]  | 0.40
.....
1  | 1  | 1  | 1    | 0   | [1.00, -0.30]  | 0.40  | [1.00, -0.30]  | 0.40
.....
=====

```

Epoch : 9, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -0.30]  | 0.40  | [1.00, -0.30]  | 0.30
.....
0  | 1  | 1  | 0    | 1   | [1.00, -0.30]  | 0.30  | [1.00, -0.20]  | 0.40
.....
1  | 0  | 1  | 1    | 0   | [1.00, -0.20]  | 0.40  | [1.00, -0.20]  | 0.40
.....
1  | 1  | 1  | 1    | 0   | [1.00, -0.20]  | 0.40  | [1.00, -0.20]  | 0.40
.....
=====

```

Epoch : 10, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -0.20]  | 0.40  | [1.00, -0.20]  | 0.30
.....
0  | 1  | 1  | 1    | 0   | [1.00, -0.20]  | 0.30  | [1.00, -0.20]  | 0.30
.....
1  | 0  | 1  | 1    | 0   | [1.00, -0.20]  | 0.30  | [1.00, -0.20]  | 0.30
.....
1  | 1  | 1  | 1    | 0   | [1.00, -0.20]  | 0.30  | [1.00, -0.20]  | 0.30
.....
=====

```

Epoch : 11, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -0.20]  | 0.30  | [1.00, -0.20]  | 0.20
.....
0  | 1  | 1  | 0    | 1   | [1.00, -0.20]  | 0.20  | [1.00, -0.10]  | 0.30
.....
1  | 0  | 1  | 1    | 0   | [1.00, -0.10]  | 0.30  | [1.00, -0.10]  | 0.30
.....
1  | 1  | 1  | 1    | 0   | [1.00, -0.10]  | 0.30  | [1.00, -0.10]  | 0.30
.....
=====

```

=====
Epoch : 12, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.10]	0.30	[1.00, -0.10]	0.20
0	1	1	1	0	[1.00, -0.10]	0.20	[1.00, -0.10]	0.20
1	0	1	1	0	[1.00, -0.10]	0.20	[1.00, -0.10]	0.20
1	1	1	1	0	[1.00, -0.10]	0.20	[1.00, -0.10]	0.20

=====
Epoch : 13, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.10]	0.20	[1.00, -0.10]	0.10
0	1	1	0	1	[1.00, -0.10]	0.10	[1.00, -0.00]	0.20
1	0	1	1	0	[1.00, -0.00]	0.20	[1.00, -0.00]	0.20
1	1	1	1	0	[1.00, -0.00]	0.20	[1.00, -0.00]	0.20

=====
Epoch : 14, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.00]	0.20	[1.00, -0.00]	0.10
0	1	1	1	0	[1.00, -0.00]	0.10	[1.00, -0.00]	0.10
1	0	1	1	0	[1.00, -0.00]	0.10	[1.00, -0.00]	0.10
1	1	1	1	0	[1.00, -0.00]	0.10	[1.00, -0.00]	0.10

=====
Epoch : 15, learning_rate : 0.1
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.00]	0.10	[1.00, -0.00]	0.00
0	1	1	0	1	[1.00, -0.00]	0.00	[1.00, 0.10]	0.10
1	0	1	1	0	[1.00, 0.10]	0.10	[1.00, 0.10]	0.10

1		1		1		1		0		[1.00, 0.10]		0.10		[1.00, 0.10]		0.10
---	--	---	--	---	--	---	--	---	--	--------------	--	------	--	--------------	--	------

Epoch : 16, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, 0.10]		0.10		[1.00, 0.10]		0.00
0		1		1		1		0		[1.00, 0.10]		0.00		[1.00, 0.10]		0.00
1		0		1		1		0		[1.00, 0.10]		0.00		[1.00, 0.10]		0.00
1		1		1		1		0		[1.00, 0.10]		0.00		[1.00, 0.10]		0.00

Epoch : 17, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, 0.10]		0.00		[1.00, 0.10]		-0.10
0		1		1		0		1		[1.00, 0.10]		-0.10		[1.00, 0.20]		0.00
1		0		1		1		0		[1.00, 0.20]		0.00		[1.00, 0.20]		0.00
1		1		1		1		0		[1.00, 0.20]		0.00		[1.00, 0.20]		0.00

Epoch : 18, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, 0.20]		0.00		[1.00, 0.20]		-0.10
0		1		1		1		0		[1.00, 0.20]		-0.10		[1.00, 0.20]		-0.10
1		0		1		1		0		[1.00, 0.20]		-0.10		[1.00, 0.20]		-0.10
1		1		1		1		0		[1.00, 0.20]		-0.10		[1.00, 0.20]		-0.10

Epoch : 19, learning_rate : 0.1

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[1.00, 0.20]		-0.10		[1.00, 0.20]		-0.10
0		1		1		1		0		[1.00, 0.20]		-0.10		[1.00, 0.20]		-0.10

1	0	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
1	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10

Epoch : 1, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.30
0	1	1	0	1	[1.00, -1.00]	0.30	[1.00, -0.80]	0.50
1	0	1	1	0	[1.00, -0.80]	0.50	[1.00, -0.80]	0.50
1	1	1	1	0	[1.00, -0.80]	0.50	[1.00, -0.80]	0.50

Epoch : 2, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.80]	0.50	[1.00, -0.80]	0.30
0	1	1	0	1	[1.00, -0.80]	0.30	[1.00, -0.60]	0.50
1	0	1	1	0	[1.00, -0.60]	0.50	[1.00, -0.60]	0.50
1	1	1	1	0	[1.00, -0.60]	0.50	[1.00, -0.60]	0.50

Epoch : 3, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.60]	0.50	[1.00, -0.60]	0.30
0	1	1	0	1	[1.00, -0.60]	0.30	[1.00, -0.40]	0.50
1	0	1	1	0	[1.00, -0.40]	0.50	[1.00, -0.40]	0.50
1	1	1	1	0	[1.00, -0.40]	0.50	[1.00, -0.40]	0.50

Epoch : 4, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.40]	0.50	[1.00, -0.40]	0.30

0	1	1	0	1	[1.00, -0.40]	0.30	[1.00, -0.20]	0.50
1	0	1	1	0	[1.00, -0.20]	0.50	[1.00, -0.20]	0.50
1	1	1	1	0	[1.00, -0.20]	0.50	[1.00, -0.20]	0.50

Epoch : 5, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.20]	0.50	[1.00, -0.20]	0.30
0	1	1	1	0	[1.00, -0.20]	0.30	[1.00, -0.20]	0.30
1	0	1	1	0	[1.00, -0.20]	0.30	[1.00, -0.20]	0.30
1	1	1	1	0	[1.00, -0.20]	0.30	[1.00, -0.20]	0.30

Epoch : 6, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.20]	0.30	[1.00, -0.20]	0.10
0	1	1	0	1	[1.00, -0.20]	0.10	[1.00, -0.00]	0.30
1	0	1	1	0	[1.00, -0.00]	0.30	[1.00, -0.00]	0.30
1	1	1	1	0	[1.00, -0.00]	0.30	[1.00, -0.00]	0.30

Epoch : 7, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.00]	0.30	[1.00, -0.00]	0.10
0	1	1	1	0	[1.00, -0.00]	0.10	[1.00, -0.00]	0.10
1	0	1	1	0	[1.00, -0.00]	0.10	[1.00, -0.00]	0.10
1	1	1	1	0	[1.00, -0.00]	0.10	[1.00, -0.00]	0.10

Epoch : 8, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

0	0	0	1	-1	[1.00, -0.00]	0.10	[1.00, -0.00]	-0.10
.....								
0	1	1	0	1	[1.00, -0.00]	-0.10	[1.00, 0.20]	0.10
.....								
1	0	1	1	0	[1.00, 0.20]	0.10	[1.00, 0.20]	0.10
.....								
1	1	1	1	0	[1.00, 0.20]	0.10	[1.00, 0.20]	0.10
.....								

Epoch : 9, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, 0.20]	0.10	[1.00, 0.20]	-0.10
.....								
0	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
.....								
1	0	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
.....								
1	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
.....								

Epoch : 10, learning_rate : 0.2

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	0	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
.....								
0	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
.....								
1	0	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
.....								
1	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
.....								

Epoch : 1, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new

0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.20
.....								
0	1	1	0	1	[1.00, -1.00]	0.20	[1.00, -0.70]	0.50
.....								
1	0	1	1	0	[1.00, -0.70]	0.50	[1.00, -0.70]	0.50
.....								
1	1	1	1	0	[1.00, -0.70]	0.50	[1.00, -0.70]	0.50
.....								

Epoch : 2, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.70]	0.50	[1.00, -0.70]	0.20
0	1	1	0	1	[1.00, -0.70]	0.20	[1.00, -0.40]	0.50
1	0	1	1	0	[1.00, -0.40]	0.50	[1.00, -0.40]	0.50
1	1	1	1	0	[1.00, -0.40]	0.50	[1.00, -0.40]	0.50

Epoch : 3, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.40]	0.50	[1.00, -0.40]	0.20
0	1	1	0	1	[1.00, -0.40]	0.20	[1.00, -0.10]	0.50
1	0	1	1	0	[1.00, -0.10]	0.50	[1.00, -0.10]	0.50
1	1	1	1	0	[1.00, -0.10]	0.50	[1.00, -0.10]	0.50

Epoch : 4, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.10]	0.50	[1.00, -0.10]	0.20
0	1	1	1	0	[1.00, -0.10]	0.20	[1.00, -0.10]	0.20
1	0	1	1	0	[1.00, -0.10]	0.20	[1.00, -0.10]	0.20
1	1	1	1	0	[1.00, -0.10]	0.20	[1.00, -0.10]	0.20

Epoch : 5, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.10]	0.20	[1.00, -0.10]	-0.10
0	1	1	0	1	[1.00, -0.10]	-0.10	[1.00, 0.20]	0.20
1	0	1	1	0	[1.00, 0.20]	0.20	[1.00, 0.20]	0.20
1	1	1	1	0	[1.00, 0.20]	0.20	[1.00, 0.20]	0.20

Epoch : 6, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, 0.20]	0.20	[1.00, 0.20]	-0.10
0	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
1	0	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
1	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10

Epoch : 7, learning_rate : 0.3

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
0	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
1	0	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10
1	1	1	1	0	[1.00, 0.20]	-0.10	[1.00, 0.20]	-0.10

Epoch : 1, learning_rate : 0.4

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.10
0	1	1	0	1	[1.00, -1.00]	0.10	[1.00, -0.60]	0.50
1	0	1	1	0	[1.00, -0.60]	0.50	[1.00, -0.60]	0.50
1	1	1	1	0	[1.00, -0.60]	0.50	[1.00, -0.60]	0.50

Epoch : 2, learning_rate : 0.4

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.60]	0.50	[1.00, -0.60]	0.10
0	1	1	0	1	[1.00, -0.60]	0.10	[1.00, -0.20]	0.50
1	0	1	1	0	[1.00, -0.20]	0.50	[1.00, -0.20]	0.50
1	1	1	1	0	[1.00, -0.20]	0.50	[1.00, -0.20]	0.50

```

.....
=====
Epoch : 3, learning_rate : 0.4
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.20] | 0.50 | [1.00, -0.20] | 0.10
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.20] | 0.10 | [1.00, 0.20] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.20] | 0.50 | [1.00, 0.20] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.20] | 0.50 | [1.00, 0.20] | 0.50
.....
=====

```

```

Epoch : 4, learning_rate : 0.4
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.20] | 0.50 | [1.00, 0.20] | 0.10
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.20] | 0.10 | [1.00, 0.20] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.20] | 0.10 | [1.00, 0.20] | 0.10
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.20] | 0.10 | [1.00, 0.20] | 0.10
.....
=====

```

```

Epoch : 5, learning_rate : 0.4
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.20] | 0.10 | [1.00, 0.20] | -0.30
.....
0 | 1 | 1 | 0 | 1 | [1.00, 0.20] | -0.30 | [1.00, 0.60] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.60] | 0.10 | [1.00, 0.60] | 0.10
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.60] | 0.10 | [1.00, 0.60] | 0.10
.....
=====

```

```

Epoch : 6, learning_rate : 0.4
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.60] | 0.10 | [1.00, 0.60] | -0.30
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.60] | -0.30 | [1.00, 0.60] | -0.30
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.60] | -0.30 | [1.00, 0.60] | -0.30
.....

```

```
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.60] | -0.30 | [1.00, 0.60] | -0.30
.....
```

```
=====
Epoch : 7, learning_rate : 0.4
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.60] | -0.30 | [1.00, 0.60] | -0.30
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.60] | -0.30 | [1.00, 0.60] | -0.30
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.60] | -0.30 | [1.00, 0.60] | -0.30
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.60] | -0.30 | [1.00, 0.60] | -0.30
.....
```

```
=====
Epoch : 1, learning_rate : 0.5
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.00
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.00 | [1.00, -0.50] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.50] | 0.50 | [1.00, -0.50] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -0.50] | 0.50 | [1.00, -0.50] | 0.50
.....
```

```
=====
Epoch : 2, learning_rate : 0.5
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.50] | 0.50 | [1.00, -0.50] | 0.00
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.50] | 0.00 | [1.00, 0.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.00] | 0.50 | [1.00, 0.00] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.00] | 0.50 | [1.00, 0.00] | 0.50
.....
```

```
=====
Epoch : 3, learning_rate : 0.5
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.00] | 0.50 | [1.00, 0.00] | 0.00
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.00] | 0.00 | [1.00, 0.00] | 0.00
.....
```

```

.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.00] | 0.00 | [1.00, 0.00] | 0.00
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.00] | 0.00 | [1.00, 0.00] | 0.00
.....

```

Epoch : 4, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.00] | 0.00 | [1.00, 0.00] | -0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, 0.00] | -0.50 | [1.00, 0.50] | 0.00
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.50] | 0.00 | [1.00, 0.50] | 0.00
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.50] | 0.00 | [1.00, 0.50] | 0.00
.....

```

Epoch : 5, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.50] | 0.00 | [1.00, 0.50] | -0.50
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....

```

Epoch : 6, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.50] | -0.50 | [1.00, 0.50] | -0.50
.....

```

Epoch : 1, learning_rate : 0.6

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | -0.10

```

```

.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | -0.10 | [1.00, -0.40] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.40] | 0.50 | [1.00, -0.40] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -0.40] | 0.50 | [1.00, -0.40] | 0.50
.....

```

=====
Epoch : 2, learning_rate : 0.6
=====

```

X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.40] | 0.50 | [1.00, -0.40] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.40] | -0.10 | [1.00, 0.20] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.20] | 0.50 | [1.00, 0.20] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.20] | 0.50 | [1.00, 0.20] | 0.50
.....

```

=====
Epoch : 3, learning_rate : 0.6
=====

```

X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.20] | 0.50 | [1.00, 0.20] | -0.10
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....

```

=====
Epoch : 4, learning_rate : 0.6
=====

```

X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.20] | -0.10 | [1.00, 0.20] | -0.10
.....

```

=====
Epoch : 1, learning_rate : 0.7
=====

```

X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new

```

```

-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | -0.20
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | -0.20 | [1.00, -0.30] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.30] | 0.50 | [1.00, -0.30] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, -0.30] | 0.50 | [1.00, -0.30] | 0.50
.....
=====

```

Epoch : 2, learning_rate : 0.7

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -0.30] | 0.50 | [1.00, -0.30] | -0.20
.....
0 | 1 | 1 | 0 | 1 | [1.00, -0.30] | -0.20 | [1.00, 0.40] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.40] | 0.50 | [1.00, 0.40] | 0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.40] | 0.50 | [1.00, 0.40] | 0.50
.....
=====

```

Epoch : 3, learning_rate : 0.7

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, 0.40] | 0.50 | [1.00, 0.40] | -0.20
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.40] | -0.20 | [1.00, 0.40] | -0.20
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.40] | -0.20 | [1.00, 0.40] | -0.20
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.40] | -0.20 | [1.00, 0.40] | -0.20
.....
=====

```

Epoch : 4, learning_rate : 0.7

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [1.00, 0.40] | -0.20 | [1.00, 0.40] | -0.20
.....
0 | 1 | 1 | 1 | 0 | [1.00, 0.40] | -0.20 | [1.00, 0.40] | -0.20
.....
1 | 0 | 1 | 1 | 0 | [1.00, 0.40] | -0.20 | [1.00, 0.40] | -0.20
.....
1 | 1 | 1 | 1 | 0 | [1.00, 0.40] | -0.20 | [1.00, 0.40] | -0.20
.....
=====

```

Epoch : 1, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.30
0	1	1	0	1	[1.00, -1.00]	-0.30	[1.00, -0.20]	0.50
1	0	1	1	0	[1.00, -0.20]	0.50	[1.00, -0.20]	0.50
1	1	1	1	0	[1.00, -0.20]	0.50	[1.00, -0.20]	0.50

Epoch : 2, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.20]	0.50	[1.00, -0.20]	-0.30
0	1	1	0	1	[1.00, -0.20]	-0.30	[1.00, 0.60]	0.50
1	0	1	1	0	[1.00, 0.60]	0.50	[1.00, 0.60]	0.50
1	1	1	1	0	[1.00, 0.60]	0.50	[1.00, 0.60]	0.50

Epoch : 3, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, 0.60]	0.50	[1.00, 0.60]	-0.30
0	1	1	1	0	[1.00, 0.60]	-0.30	[1.00, 0.60]	-0.30
1	0	1	1	0	[1.00, 0.60]	-0.30	[1.00, 0.60]	-0.30
1	1	1	1	0	[1.00, 0.60]	-0.30	[1.00, 0.60]	-0.30

Epoch : 4, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.60]	-0.30	[1.00, 0.60]	-0.30
0	1	1	1	0	[1.00, 0.60]	-0.30	[1.00, 0.60]	-0.30
1	0	1	1	0	[1.00, 0.60]	-0.30	[1.00, 0.60]	-0.30
1	1	1	1	0	[1.00, 0.60]	-0.30	[1.00, 0.60]	-0.30

=====
Epoch : 1, learning_rate : 0.9
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.40
0	1	1	0	1	[1.00, -1.00]	-0.40	[1.00, -0.10]	0.50
1	0	1	1	0	[1.00, -0.10]	0.50	[1.00, -0.10]	0.50
1	1	1	1	0	[1.00, -0.10]	0.50	[1.00, -0.10]	0.50

.....

=====
Epoch : 2, learning_rate : 0.9
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -0.10]	0.50	[1.00, -0.10]	-0.40
0	1	1	0	1	[1.00, -0.10]	-0.40	[1.00, 0.80]	0.50
1	0	1	1	0	[1.00, 0.80]	0.50	[1.00, 0.80]	0.50
1	1	1	1	0	[1.00, 0.80]	0.50	[1.00, 0.80]	0.50

.....

=====
Epoch : 3, learning_rate : 0.9
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, 0.80]	0.50	[1.00, 0.80]	-0.40
0	1	1	1	0	[1.00, 0.80]	-0.40	[1.00, 0.80]	-0.40
1	0	1	1	0	[1.00, 0.80]	-0.40	[1.00, 0.80]	-0.40
1	1	1	1	0	[1.00, 0.80]	-0.40	[1.00, 0.80]	-0.40

.....

=====
Epoch : 4, learning_rate : 0.9
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[1.00, 0.80]	-0.40	[1.00, 0.80]	-0.40
0	1	1	1	0	[1.00, 0.80]	-0.40	[1.00, 0.80]	-0.40
1	0	1	1	0	[1.00, 0.80]	-0.40	[1.00, 0.80]	-0.40

.....

1		1		1		1		0		[1.00, 0.80]		-0.40		[1.00, 0.80]		-0.40
---	--	---	--	---	--	---	--	---	--	--------------	--	-------	--	--------------	--	-------

Epoch : 1, learning_rate : 1.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, -1.00]		0.50		[1.00, -1.00]		-0.50
0		1		1		0		1		[1.00, -1.00]		-0.50		[1.00, 0.00]		0.50
1		0		1		1		0		[1.00, 0.00]		0.50		[1.00, 0.00]		0.50
1		1		1		1		0		[1.00, 0.00]		0.50		[1.00, 0.00]		0.50

Epoch : 2, learning_rate : 1.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, 0.00]		0.50		[1.00, 0.00]		-0.50
0		1		1		0		1		[1.00, 0.00]		-0.50		[1.00, 1.00]		0.50
1		0		1		1		0		[1.00, 1.00]		0.50		[1.00, 1.00]		0.50
1		1		1		1		0		[1.00, 1.00]		0.50		[1.00, 1.00]		0.50

Epoch : 3, learning_rate : 1.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[1.00, 1.00]		0.50		[1.00, 1.00]		-0.50
0		1		1		1		0		[1.00, 1.00]		-0.50		[1.00, 1.00]		-0.50
1		0		1		1		0		[1.00, 1.00]		-0.50		[1.00, 1.00]		-0.50
1		1		1		1		0		[1.00, 1.00]		-0.50		[1.00, 1.00]		-0.50

Epoch : 4, learning_rate : 1.0

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[1.00, 1.00]		-0.50		[1.00, 1.00]		-0.50
0		1		1		1		0		[1.00, 1.00]		-0.50		[1.00, 1.00]		-0.50

```

1 | 0 | 1 | 1 | 0 | [1.00, 1.00] | -0.50 | [1.00, 1.00] | -0.50
.....
1 | 1 | 1 | 1 | 0 | [1.00, 1.00] | -0.50 | [1.00, 1.00] | -0.50
.....
=====

```

OR Gate summery

LR	Converged	Epochs	Updates	Weights	Bias	Margin
0.0	False	200	400	[1.00, -1.00]	0.50	0.35
0.1	True	19	30	[1.00, 0.20]	-0.10	0.10
0.2	True	10	15	[1.00, 0.20]	-0.10	0.10
0.3	True	7	10	[1.00, 0.20]	-0.10	0.10
0.4	True	7	10	[1.00, 0.60]	-0.30	0.26
0.5	True	6	8	[1.00, 0.50]	-0.50	0.00
0.6	True	4	5	[1.00, 0.20]	-0.10	0.10
0.7	True	4	5	[1.00, 0.40]	-0.20	0.19
0.8	True	4	5	[1.00, 0.60]	-0.30	0.26
0.9	True	4	5	[1.00, 0.80]	-0.40	0.31
1.0	True	4	5	[1.00, 1.00]	-0.50	0.35

Epoch : 1, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
-> [Cycle Detected] Period: 1 updates
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 2, learning_rate : 0.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.50
.....
=====

```

Epoch : 3, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 4, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 5, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 6, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 7, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 8, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 9, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

Epoch : 10, learning_rate : 0.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
0	1	1	0	1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	0	1	1	0	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50
1	1	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.50

```

.....
=====
Epoch : 1, learning_rate : 0.1
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.40
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.40  | [1.00, -0.90]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -0.90]  | 0.50  | [1.00, -0.90]  | 0.50
.....
1  | 1  | 0  | 1    | -1  | [1.00, -0.90]  | 0.50  | [0.90, -1.00]  | 0.40
.....
=====

```

```

Epoch : 2, learning_rate : 0.1
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [0.90, -1.00]  | 0.40  | [0.90, -1.00]  | 0.30
.....
0  | 1  | 1  | 0    | 1   | [0.90, -1.00]  | 0.30  | [0.90, -0.90]  | 0.40
.....
1  | 0  | 1  | 1    | 0   | [0.90, -0.90]  | 0.40  | [0.90, -0.90]  | 0.40
.....
1  | 1  | 0  | 1    | -1  | [0.90, -0.90]  | 0.40  | [0.80, -1.00]  | 0.30
.....
=====

```

```

Epoch : 3, learning_rate : 0.1
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [0.80, -1.00]  | 0.30  | [0.80, -1.00]  | 0.20
.....
0  | 1  | 1  | 0    | 1   | [0.80, -1.00]  | 0.20  | [0.80, -0.90]  | 0.30
.....
1  | 0  | 1  | 1    | 0   | [0.80, -0.90]  | 0.30  | [0.80, -0.90]  | 0.30
.....
1  | 1  | 0  | 1    | -1  | [0.80, -0.90]  | 0.30  | [0.70, -1.00]  | 0.20
.....
=====

```

```

Epoch : 4, learning_rate : 0.1
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [0.70, -1.00]  | 0.20  | [0.70, -1.00]  | 0.10
.....
0  | 1  | 1  | 0    | 1   | [0.70, -1.00]  | 0.10  | [0.70, -0.90]  | 0.20
.....
1  | 0  | 1  | 1    | 0   | [0.70, -0.90]  | 0.20  | [0.70, -0.90]  | 0.20
.....

```

```
.....
1 | 1 | 0 | 1 | -1 | [0.70, -0.90] | 0.20 | [0.60, -1.00] | 0.10
.....
```

```
=====
Epoch : 5, learning_rate : 0.1
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.60, -1.00] | 0.10 | [0.60, -1.00] | 0.00
.....
0 | 1 | 1 | 0 | 1 | [0.60, -1.00] | 0.00 | [0.60, -0.90] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.60, -0.90] | 0.10 | [0.60, -0.90] | 0.10
.....
1 | 1 | 0 | 0 | 0 | [0.60, -0.90] | 0.10 | [0.60, -0.90] | 0.10
.....
```

```
=====
Epoch : 6, learning_rate : 0.1
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.60, -0.90] | 0.10 | [0.60, -0.90] | 0.00
.....
0 | 1 | 1 | 0 | 1 | [0.60, -0.90] | 0.00 | [0.60, -0.80] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.60, -0.80] | 0.10 | [0.60, -0.80] | 0.10
.....
1 | 1 | 0 | 0 | 0 | [0.60, -0.80] | 0.10 | [0.60, -0.80] | 0.10
.....
```

```
=====
Epoch : 7, learning_rate : 0.1
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.60, -0.80] | 0.10 | [0.60, -0.80] | 0.00
.....
0 | 1 | 1 | 0 | 1 | [0.60, -0.80] | 0.00 | [0.60, -0.70] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.60, -0.70] | 0.10 | [0.60, -0.70] | 0.10
.....
1 | 1 | 0 | 1 | -1 | [0.60, -0.70] | 0.10 | [0.50, -0.80] | 0.00
.....
```

```
=====
Epoch : 8, learning_rate : 0.1
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.50, -0.80] | 0.00 | [0.50, -0.80] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [0.50, -0.80] | -0.10 | [0.50, -0.70] | 0.00
.....
```

```

.....
1 | 0 | 1 | 1 | 0 | [0.50, -0.70] | 0.00 | [0.50, -0.70] | 0.00
.....
1 | 1 | 0 | 0 | 0 | [0.50, -0.70] | 0.00 | [0.50, -0.70] | 0.00
.....

```

Epoch : 9, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.50, -0.70] | 0.00 | [0.50, -0.70] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [0.50, -0.70] | -0.10 | [0.50, -0.60] | 0.00
.....
1 | 0 | 1 | 1 | 0 | [0.50, -0.60] | 0.00 | [0.50, -0.60] | 0.00
.....
1 | 1 | 0 | 0 | 0 | [0.50, -0.60] | 0.00 | [0.50, -0.60] | 0.00
.....

```

Epoch : 10, learning_rate : 0.1

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.50, -0.60] | 0.00 | [0.50, -0.60] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [0.50, -0.60] | -0.10 | [0.50, -0.50] | 0.00
.....
1 | 0 | 1 | 1 | 0 | [0.50, -0.50] | 0.00 | [0.50, -0.50] | 0.00
.....
1 | 1 | 0 | 1 | -1 | [0.50, -0.50] | 0.00 | [0.40, -0.60] | -0.10
.....

```

Epoch : 1, learning_rate : 0.2

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.30
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.30 | [1.00, -0.80] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.80] | 0.50 | [1.00, -0.80] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [1.00, -0.80] | 0.50 | [0.80, -1.00] | 0.30
.....

```

Epoch : 2, learning_rate : 0.2

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.80, -1.00] | 0.30 | [0.80, -1.00] | 0.10

```

```

.....
0 | 1 | 1 | 0 | 1 | [0.80, -1.00] | 0.10 | [0.80, -0.80] | 0.30
.....
1 | 0 | 1 | 1 | 0 | [0.80, -0.80] | 0.30 | [0.80, -0.80] | 0.30
.....
1 | 1 | 0 | 1 | -1 | [0.80, -0.80] | 0.30 | [0.60, -1.00] | 0.10
.....

```

=====
Epoch : 3, learning_rate : 0.2
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[0.60, -1.00]	0.10	[0.60, -1.00]	-0.10
0	1	1	0	1	[0.60, -1.00]	-0.10	[0.60, -0.80]	0.10
1	0	1	1	0	[0.60, -0.80]	0.10	[0.60, -0.80]	0.10
1	1	0	0	0	[0.60, -0.80]	0.10	[0.60, -0.80]	0.10

=====
Epoch : 4, learning_rate : 0.2
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[0.60, -0.80]	0.10	[0.60, -0.80]	-0.10
0	1	1	0	1	[0.60, -0.80]	-0.10	[0.60, -0.60]	0.10
1	0	1	1	0	[0.60, -0.60]	0.10	[0.60, -0.60]	0.10
1	1	0	1	-1	[0.60, -0.60]	0.10	[0.40, -0.80]	-0.10

=====
Epoch : 5, learning_rate : 0.2
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.40, -0.80]	-0.10	[0.40, -0.80]	-0.10
0	1	1	0	1	[0.40, -0.80]	-0.10	[0.40, -0.60]	0.10
1	0	1	1	0	[0.40, -0.60]	0.10	[0.40, -0.60]	0.10
1	1	0	0	0	[0.40, -0.60]	0.10	[0.40, -0.60]	0.10

=====
Epoch : 6, learning_rate : 0.2
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [0.40, -0.60] | 0.10 | [0.40, -0.60] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [0.40, -0.60] | -0.10 | [0.40, -0.40] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.40, -0.40] | 0.10 | [0.40, -0.40] | 0.10
.....
1 | 1 | 0 | 1 | -1 | [0.40, -0.40] | 0.10 | [0.20, -0.60] | -0.10
.....
=====

```

Epoch : 7, learning_rate : 0.2

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [0.20, -0.60] | -0.10 | [0.20, -0.60] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [0.20, -0.60] | -0.10 | [0.20, -0.40] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.20, -0.40] | 0.10 | [0.20, -0.40] | 0.10
.....
1 | 1 | 0 | 0 | 0 | [0.20, -0.40] | 0.10 | [0.20, -0.40] | 0.10
.....
=====

```

Epoch : 8, learning_rate : 0.2

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.20, -0.40] | 0.10 | [0.20, -0.40] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [0.20, -0.40] | -0.10 | [0.20, -0.20] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.20, -0.20] | 0.10 | [0.20, -0.20] | 0.10
.....
1 | 1 | 0 | 1 | -1 | [0.20, -0.20] | 0.10 | [0.00, -0.40] | -0.10
.....
=====

```

Epoch : 9, learning_rate : 0.2

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 0 | 0 | [0.00, -0.40] | -0.10 | [0.00, -0.40] | -0.10
.....
0 | 1 | 1 | 0 | 1 | [0.00, -0.40] | -0.10 | [0.00, -0.20] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.00, -0.20] | 0.10 | [0.00, -0.20] | 0.10
.....
1 | 1 | 0 | 0 | 0 | [0.00, -0.20] | 0.10 | [0.00, -0.20] | 0.10
.....
=====

```

Epoch : 10, learning_rate : 0.2

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [0.00, -0.20]  | 0.10  | [0.00, -0.20]  | -0.10
.....
0  | 1  | 1  | 0    | 1   | [0.00, -0.20]  | -0.10 | [0.00, -0.00]  | 0.10
.....
1  | 0  | 1  | 1    | 0   | [0.00, -0.00]  | 0.10  | [0.00, -0.00]  | 0.10
.....
1  | 1  | 0  | 1    | -1  | [0.00, -0.00]  | 0.10  | [-0.20, -0.20] | -0.10
=====

```

```

.....
=====

```

Epoch : 1, learning_rate : 0.3

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [1.00, -1.00]  | 0.50  | [1.00, -1.00]  | 0.20
.....
0  | 1  | 1  | 0    | 1   | [1.00, -1.00]  | 0.20  | [1.00, -0.70]  | 0.50
.....
1  | 0  | 1  | 1    | 0   | [1.00, -0.70]  | 0.50  | [1.00, -0.70]  | 0.50
.....
1  | 1  | 0  | 1    | -1  | [1.00, -0.70]  | 0.50  | [0.70, -1.00]  | 0.20
=====

```

Epoch : 2, learning_rate : 0.3

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [0.70, -1.00]  | 0.20  | [0.70, -1.00]  | -0.10
.....
0  | 1  | 1  | 0    | 1   | [0.70, -1.00]  | -0.10 | [0.70, -0.70]  | 0.20
.....
1  | 0  | 1  | 1    | 0   | [0.70, -0.70]  | 0.20  | [0.70, -0.70]  | 0.20
.....
1  | 1  | 0  | 1    | -1  | [0.70, -0.70]  | 0.20  | [0.40, -1.00]  | -0.10
=====

```

Epoch : 3, learning_rate : 0.3

```

=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 0    | 0   | [0.40, -1.00]  | -0.10 | [0.40, -1.00]  | -0.10
.....
0  | 1  | 1  | 0    | 1   | [0.40, -1.00]  | -0.10 | [0.40, -0.70]  | 0.20
.....
1  | 0  | 1  | 1    | 0   | [0.40, -0.70]  | 0.20  | [0.40, -0.70]  | 0.20
.....
1  | 1  | 0  | 0    | 0   | [0.40, -0.70]  | 0.20  | [0.40, -0.70]  | 0.20
=====

```

```

.....
=====
Epoch : 4, learning_rate : 0.3
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [0.40, -0.70]  | 0.20  | [0.40, -0.70]  | -0.10
.....
0  | 1  | 1  | 0    | 1   | [0.40, -0.70]  | -0.10 | [0.40, -0.40]  | 0.20
.....
1  | 0  | 1  | 1    | 0   | [0.40, -0.40]  | 0.20  | [0.40, -0.40]  | 0.20
.....
1  | 1  | 0  | 1    | -1  | [0.40, -0.40]  | 0.20  | [0.10, -0.70]  | -0.10
.....
=====

```

```

Epoch : 5, learning_rate : 0.3
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 0    | 0   | [0.10, -0.70]  | -0.10 | [0.10, -0.70]  | -0.10
.....
0  | 1  | 1  | 0    | 1   | [0.10, -0.70]  | -0.10 | [0.10, -0.40]  | 0.20
.....
1  | 0  | 1  | 1    | 0   | [0.10, -0.40]  | 0.20  | [0.10, -0.40]  | 0.20
.....
1  | 1  | 0  | 0    | 0   | [0.10, -0.40]  | 0.20  | [0.10, -0.40]  | 0.20
.....
=====

```

```

Epoch : 6, learning_rate : 0.3
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 1    | -1  | [0.10, -0.40]  | 0.20  | [0.10, -0.40]  | -0.10
.....
0  | 1  | 1  | 0    | 1   | [0.10, -0.40]  | -0.10 | [0.10, -0.10]  | 0.20
.....
1  | 0  | 1  | 1    | 0   | [0.10, -0.10]  | 0.20  | [0.10, -0.10]  | 0.20
.....
1  | 1  | 0  | 1    | -1  | [0.10, -0.10]  | 0.20  | [-0.20, -0.40] | -0.10
.....
=====

```

```

Epoch : 7, learning_rate : 0.3
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0  | 0  | 0  | 0    | 0   | [-0.20, -0.40] | -0.10 | [-0.20, -0.40] |
-0.10
.....
0  | 1  | 1  | 0    | 1   | [-0.20, -0.40] | -0.10 | [-0.20, -0.10] | 0.20
.....

```

```
.....
1 | 0 | 1 | 1 | 0 | [-0.20, -0.10] | 0.20 | [-0.20, -0.10] | 0.20
```

```
.....
1 | 1 | 0 | 0 | 0 | [-0.20, -0.10] | 0.20 | [-0.20, -0.10] | 0.20
```

```
.....
=====
Epoch : 8, learning_rate : 0.3
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
```

```
-----
0 | 0 | 0 | 1 | -1 | [-0.20, -0.10] | 0.20 | [-0.20, -0.10] |
-0.10
```

```
.....
0 | 1 | 1 | 0 | 1 | [-0.20, -0.10] | -0.10 | [-0.20, 0.20] | 0.20
```

```
.....
1 | 0 | 1 | 1 | 0 | [-0.20, 0.20] | 0.20 | [-0.20, 0.20] | 0.20
```

```
.....
1 | 1 | 0 | 1 | -1 | [-0.20, 0.20] | 0.20 | [-0.50, -0.10] | -0.10
```

```
.....
=====
Epoch : 9, learning_rate : 0.3
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
```

```
-----
0 | 0 | 0 | 0 | 0 | [-0.50, -0.10] | -0.10 | [-0.50, -0.10] |
-0.10
```

```
.....
0 | 1 | 1 | 0 | 1 | [-0.50, -0.10] | -0.10 | [-0.50, 0.20] | 0.20
```

```
.....
1 | 0 | 1 | 0 | 1 | [-0.50, 0.20] | 0.20 | [-0.20, 0.20] | 0.50
```

```
.....
1 | 1 | 0 | 1 | -1 | [-0.20, 0.20] | 0.50 | [-0.50, -0.10] | 0.20
```

```
.....
=====
Epoch : 10, learning_rate : 0.3
=====
```

```
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
```

```
-----
-> [Cycle Detected] Period: 4 updates
0 | 0 | 0 | 1 | -1 | [-0.50, -0.10] | 0.20 | [-0.50, -0.10] |
-0.10
```

```
.....
0 | 1 | 1 | 0 | 1 | [-0.50, -0.10] | -0.10 | [-0.50, 0.20] | 0.20
```

```

.....
1 | 0 | 1 | 0 | 1 | [-0.50, 0.20] | 0.20 | [-0.20, 0.20] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [-0.20, 0.20] | 0.50 | [-0.50, -0.10] | 0.20
.....

```

```

=====
Epoch : 1, learning_rate : 0.4
=====

```

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	0.10
0	1	1	0	1	[1.00, -1.00]	0.10	[1.00, -0.60]	0.50
1	0	1	1	0	[1.00, -0.60]	0.50	[1.00, -0.60]	0.50
1	1	0	1	-1	[1.00, -0.60]	0.50	[0.60, -1.00]	0.10

```

=====
Epoch : 2, learning_rate : 0.4
=====

```

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[0.60, -1.00]	0.10	[0.60, -1.00]	-0.30
0	1	1	0	1	[0.60, -1.00]	-0.30	[0.60, -0.60]	0.10
1	0	1	1	0	[0.60, -0.60]	0.10	[0.60, -0.60]	0.10
1	1	0	1	-1	[0.60, -0.60]	0.10	[0.20, -1.00]	-0.30

```

=====
Epoch : 3, learning_rate : 0.4
=====

```

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.20, -1.00]	-0.30	[0.20, -1.00]	-0.30
0	1	1	0	1	[0.20, -1.00]	-0.30	[0.20, -0.60]	0.10
1	0	1	1	0	[0.20, -0.60]	0.10	[0.20, -0.60]	0.10
1	1	0	0	0	[0.20, -0.60]	0.10	[0.20, -0.60]	0.10

```

=====
Epoch : 4, learning_rate : 0.4
=====

```

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [0.20, -0.60] | 0.10 | [0.20, -0.60] | -0.30
.....
0 | 1 | 1 | 0 | 1 | [0.20, -0.60] | -0.30 | [0.20, -0.20] | 0.10
.....
1 | 0 | 1 | 1 | 0 | [0.20, -0.20] | 0.10 | [0.20, -0.20] | 0.10
.....
1 | 1 | 0 | 1 | -1 | [0.20, -0.20] | 0.10 | [-0.20, -0.60] | -0.30
.....

```

```

=====
Epoch : 5, learning_rate : 0.4
=====

```

```

-----
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [-0.20, -0.60] | -0.30 | [-0.20, -0.60] |
-0.30
.....
0 | 1 | 1 | 0 | 1 | [-0.20, -0.60] | -0.30 | [-0.20, -0.20] | 0.10
.....
1 | 0 | 1 | 0 | 1 | [-0.20, -0.20] | 0.10 | [0.20, -0.20] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [0.20, -0.20] | 0.50 | [-0.20, -0.60] | 0.10
.....

```

```

=====
Epoch : 6, learning_rate : 0.4
=====

```

```

-----
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
-> [Cycle Detected] Period: 4 updates
0 | 0 | 0 | 1 | -1 | [-0.20, -0.60] | 0.10 | [-0.20, -0.60] |
-0.30
.....
0 | 1 | 1 | 0 | 1 | [-0.20, -0.60] | -0.30 | [-0.20, -0.20] | 0.10
.....
1 | 0 | 1 | 0 | 1 | [-0.20, -0.20] | 0.10 | [0.20, -0.20] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [0.20, -0.20] | 0.50 | [-0.20, -0.60] | 0.10
.....

```

```

=====
Epoch : 7, learning_rate : 0.4
=====

```

```

-----
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----

```

0	0	0	1	-1	[-0.20, -0.60]	0.10	[-0.20, -0.60]	-0.30
0	1	1	0	1	[-0.20, -0.60]	-0.30	[-0.20, -0.20]	0.10
1	0	1	0	1	[-0.20, -0.20]	0.10	[0.20, -0.20]	0.50
1	1	0	1	-1	[0.20, -0.20]	0.50	[-0.20, -0.60]	0.10

Epoch : 8, learning_rate : 0.4

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.20, -0.60]	0.10	[-0.20, -0.60]	-0.30
0	1	1	0	1	[-0.20, -0.60]	-0.30	[-0.20, -0.20]	0.10
1	0	1	0	1	[-0.20, -0.20]	0.10	[0.20, -0.20]	0.50
1	1	0	1	-1	[0.20, -0.20]	0.50	[-0.20, -0.60]	0.10

Epoch : 9, learning_rate : 0.4

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.20, -0.60]	0.10	[-0.20, -0.60]	-0.30
0	1	1	0	1	[-0.20, -0.60]	-0.30	[-0.20, -0.20]	0.10
1	0	1	0	1	[-0.20, -0.20]	0.10	[0.20, -0.20]	0.50
1	1	0	1	-1	[0.20, -0.20]	0.50	[-0.20, -0.60]	0.10

Epoch : 10, learning_rate : 0.4

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [-0.20, -0.60] | 0.10 | [-0.20, -0.60] |
-0.30
.....
0 | 1 | 1 | 0 | 1 | [-0.20, -0.60] | -0.30 | [-0.20, -0.20] | 0.10
.....
1 | 0 | 1 | 0 | 1 | [-0.20, -0.20] | 0.10 | [0.20, -0.20] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [0.20, -0.20] | 0.50 | [-0.20, -0.60] | 0.10
.....

```

Epoch : 1, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | 0.00
.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | 0.00 | [1.00, -0.50] | 0.50
.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.50] | 0.50 | [1.00, -0.50] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [1.00, -0.50] | 0.50 | [0.50, -1.00] | 0.00
.....

```

Epoch : 2, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [0.50, -1.00] | 0.00 | [0.50, -1.00] | -0.50
.....
0 | 1 | 1 | 0 | 1 | [0.50, -1.00] | -0.50 | [0.50, -0.50] | 0.00
.....
1 | 0 | 1 | 1 | 0 | [0.50, -0.50] | 0.00 | [0.50, -0.50] | 0.00
.....
1 | 1 | 0 | 1 | -1 | [0.50, -0.50] | 0.00 | [0.00, -1.00] | -0.50
.....

```

Epoch : 3, learning_rate : 0.5

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 0 | 0 | [0.00, -1.00] | -0.50 | [0.00, -1.00] | -0.50
.....
0 | 1 | 1 | 0 | 1 | [0.00, -1.00] | -0.50 | [0.00, -0.50] | 0.00
.....
1 | 0 | 1 | 1 | 0 | [0.00, -0.50] | 0.00 | [0.00, -0.50] | 0.00
.....

```

1		1		0		0		0		[0.00, -0.50]		0.00		[0.00, -0.50]		0.00
---	--	---	--	---	--	---	--	---	--	---------------	--	------	--	---------------	--	------

Epoch : 4, learning_rate : 0.5

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		1		-1		[0.00, -0.50]		0.00		[0.00, -0.50]		-0.50
0		1		1		0		1		[0.00, -0.50]		-0.50		[0.00, 0.00]		0.00
1		0		1		1		0		[0.00, 0.00]		0.00		[0.00, 0.00]		0.00
1		1		0		1		-1		[0.00, 0.00]		0.00		[-0.50, -0.50]		-0.50

Epoch : 5, learning_rate : 0.5

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
0		0		0		0		0		[-0.50, -0.50]		-0.50		[-0.50, -0.50]		-0.50
0		1		1		0		1		[-0.50, -0.50]		-0.50		[-0.50, 0.00]		0.00
1		0		1		0		1		[-0.50, 0.00]		0.00		[0.00, 0.00]		0.50
1		1		0		1		-1		[0.00, 0.00]		0.50		[-0.50, -0.50]		0.00

Epoch : 6, learning_rate : 0.5

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
-> [Cycle Detected] Period: 4 updates																
0		0		0		1		-1		[-0.50, -0.50]		0.00		[-0.50, -0.50]		-0.50
0		1		1		0		1		[-0.50, -0.50]		-0.50		[-0.50, 0.00]		0.00
1		0		1		0		1		[-0.50, 0.00]		0.00		[0.00, 0.00]		0.50
1		1		0		1		-1		[0.00, 0.00]		0.50		[-0.50, -0.50]		0.00

Epoch : 7, learning_rate : 0.5

X1		X2		Ac		Pred		Err		W_old		b_old		W_new		b_new
----	--	----	--	----	--	------	--	-----	--	-------	--	-------	--	-------	--	-------

```

-----
0 | 0 | 0 | 1 | -1 | [-0.50, -0.50] | 0.00 | [-0.50, -0.50] |
-0.50
.....
0 | 1 | 1 | 0 | 1 | [-0.50, -0.50] | -0.50 | [-0.50, 0.00] | 0.00

.....
1 | 0 | 1 | 0 | 1 | [-0.50, 0.00] | 0.00 | [0.00, 0.00] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 0.50 | [-0.50, -0.50] | 0.00
.....

```

=====
Epoch : 8, learning_rate : 0.5
=====

```

-----
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [-0.50, -0.50] | 0.00 | [-0.50, -0.50] |
-0.50
.....
0 | 1 | 1 | 0 | 1 | [-0.50, -0.50] | -0.50 | [-0.50, 0.00] | 0.00

.....
1 | 0 | 1 | 0 | 1 | [-0.50, 0.00] | 0.00 | [0.00, 0.00] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 0.50 | [-0.50, -0.50] | 0.00
.....

```

=====
Epoch : 9, learning_rate : 0.5
=====

```

-----
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [-0.50, -0.50] | 0.00 | [-0.50, -0.50] |
-0.50
.....
0 | 1 | 1 | 0 | 1 | [-0.50, -0.50] | -0.50 | [-0.50, 0.00] | 0.00

.....
1 | 0 | 1 | 0 | 1 | [-0.50, 0.00] | 0.00 | [0.00, 0.00] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 0.50 | [-0.50, -0.50] | 0.00
.....

```

=====
Epoch : 10, learning_rate : 0.5
=====

```

-----
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [-0.50, -0.50] | 0.00 | [-0.50, -0.50] |
-0.50
.....
0 | 1 | 1 | 0 | 1 | [-0.50, -0.50] | -0.50 | [-0.50, 0.00] | 0.00

```

```

.....
1 | 0 | 1 | 0 | 1 | [-0.50, 0.00] | 0.00 | [0.00, 0.00] | 0.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 0.50 | [-0.50, -0.50] | 0.00
.....

```

=====
Epoch : 1, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.10
0	1	1	0	1	[1.00, -1.00]	-0.10	[1.00, -0.40]	0.50
1	0	1	1	0	[1.00, -0.40]	0.50	[1.00, -0.40]	0.50
1	1	0	1	-1	[1.00, -0.40]	0.50	[0.40, -1.00]	-0.10

=====
Epoch : 2, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.40, -1.00]	-0.10	[0.40, -1.00]	-0.10
0	1	1	0	1	[0.40, -1.00]	-0.10	[0.40, -0.40]	0.50
1	0	1	1	0	[0.40, -0.40]	0.50	[0.40, -0.40]	0.50
1	1	0	1	-1	[0.40, -0.40]	0.50	[-0.20, -1.00]	-0.10

=====
Epoch : 3, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[-0.20, -1.00]	-0.10	[-0.20, -1.00]	-0.10
0	1	1	0	1	[-0.20, -1.00]	-0.10	[-0.20, -0.40]	0.50
1	0	1	1	0	[-0.20, -0.40]	0.50	[-0.20, -0.40]	0.50
1	1	0	0	0	[-0.20, -0.40]	0.50	[-0.20, -0.40]	0.50

=====
Epoch : 4, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.20, -0.40]	0.50	[-0.20, -0.40]	-0.10
0	1	1	0	1	[-0.20, -0.40]	-0.10	[-0.20, 0.20]	0.50
1	0	1	1	0	[-0.20, 0.20]	0.50	[-0.20, 0.20]	0.50
1	1	0	1	-1	[-0.20, 0.20]	0.50	[-0.80, -0.40]	-0.10

=====
Epoch : 5, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[-0.80, -0.40]	-0.10	[-0.80, -0.40]	-0.10
0	1	1	0	1	[-0.80, -0.40]	-0.10	[-0.80, 0.20]	0.50
1	0	1	0	1	[-0.80, 0.20]	0.50	[-0.20, 0.20]	1.10
1	1	0	1	-1	[-0.20, 0.20]	1.10	[-0.80, -0.40]	0.50

=====
Epoch : 6, learning_rate : 0.6
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.80, -0.40]	0.50	[-0.80, -0.40]	-0.10
0	1	1	0	1	[-0.80, -0.40]	-0.10	[-0.80, 0.20]	0.50
-> [Cycle Detected] Period: 4 updates								
1	0	1	0	1	[-0.80, 0.20]	0.50	[-0.20, 0.20]	1.10
1	1	0	1	-1	[-0.20, 0.20]	1.10	[-0.80, -0.40]	0.50

Epoch : 7, learning_rate : 0.6

```
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [-0.80, -0.40] | 0.50 | [-0.80, -0.40] |
-0.10
.....
0 | 1 | 1 | 0    | 1  | [-0.80, -0.40] | -0.10 | [-0.80, 0.20]  | 0.50

.....
1 | 0 | 1 | 0    | 1  | [-0.80, 0.20]  | 0.50 | [-0.20, 0.20]  | 1.10
.....
1 | 1 | 0 | 1    | -1 | [-0.20, 0.20]  | 1.10 | [-0.80, -0.40] | 0.50
.....
```

Epoch : 8, learning_rate : 0.6

```
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [-0.80, -0.40] | 0.50 | [-0.80, -0.40] |
-0.10
.....
0 | 1 | 1 | 0    | 1  | [-0.80, -0.40] | -0.10 | [-0.80, 0.20]  | 0.50

.....
1 | 0 | 1 | 0    | 1  | [-0.80, 0.20]  | 0.50 | [-0.20, 0.20]  | 1.10
.....
1 | 1 | 0 | 1    | -1 | [-0.20, 0.20]  | 1.10 | [-0.80, -0.40] | 0.50
.....
```

Epoch : 9, learning_rate : 0.6

```
=====
X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1    | -1 | [-0.80, -0.40] | 0.50 | [-0.80, -0.40] |
-0.10
.....
0 | 1 | 1 | 0    | 1  | [-0.80, -0.40] | -0.10 | [-0.80, 0.20]  | 0.50

.....
1 | 0 | 1 | 0    | 1  | [-0.80, 0.20]  | 0.50 | [-0.20, 0.20]  | 1.10
.....
1 | 1 | 0 | 1    | -1 | [-0.20, 0.20]  | 1.10 | [-0.80, -0.40] | 0.50
.....
```

Epoch : 10, learning_rate : 0.6

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.80, -0.40]	0.50	[-0.80, -0.40]	-0.10
0	1	1	0	1	[-0.80, -0.40]	-0.10	[-0.80, 0.20]	0.50
1	0	1	0	1	[-0.80, 0.20]	0.50	[-0.20, 0.20]	1.10
1	1	0	1	-1	[-0.20, 0.20]	1.10	[-0.80, -0.40]	0.50

Epoch : 1, learning_rate : 0.7

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.20
0	1	1	0	1	[1.00, -1.00]	-0.20	[1.00, -0.30]	0.50
1	0	1	1	0	[1.00, -0.30]	0.50	[1.00, -0.30]	0.50
1	1	0	1	-1	[1.00, -0.30]	0.50	[0.30, -1.00]	-0.20

Epoch : 2, learning_rate : 0.7

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.30, -1.00]	-0.20	[0.30, -1.00]	-0.20
0	1	1	0	1	[0.30, -1.00]	-0.20	[0.30, -0.30]	0.50
1	0	1	1	0	[0.30, -0.30]	0.50	[0.30, -0.30]	0.50
1	1	0	1	-1	[0.30, -0.30]	0.50	[-0.40, -1.00]	-0.20

Epoch : 3, learning_rate : 0.7

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[-0.40, -1.00]	-0.20	[-0.40, -1.00]	-0.20
0	1	1	0	1	[-0.40, -1.00]	-0.20	[-0.40, -0.30]	0.50

```
.....
1 | 0 | 1 | 1 | 0 | [-0.40, -0.30] | 0.50 | [-0.40, -0.30] | 0.50
```

```
.....
1 | 1 | 0 | 0 | 0 | [-0.40, -0.30] | 0.50 | [-0.40, -0.30] | 0.50
```

```
.....
=====
Epoch : 4, learning_rate : 0.7
```

```
=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
```

```
-----
0 | 0 | 0 | 1 | -1 | [-0.40, -0.30] | 0.50 | [-0.40, -0.30] |
-0.20
```

```
.....
0 | 1 | 1 | 0 | 1 | [-0.40, -0.30] | -0.20 | [-0.40, 0.40] | 0.50
```

```
.....
1 | 0 | 1 | 1 | 0 | [-0.40, 0.40] | 0.50 | [-0.40, 0.40] | 0.50
```

```
.....
1 | 1 | 0 | 1 | -1 | [-0.40, 0.40] | 0.50 | [-1.10, -0.30] | -0.20
```

```
.....
=====
Epoch : 5, learning_rate : 0.7
```

```
=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
```

```
-----
0 | 0 | 0 | 0 | 0 | [-1.10, -0.30] | -0.20 | [-1.10, -0.30] |
-0.20
```

```
.....
0 | 1 | 1 | 0 | 1 | [-1.10, -0.30] | -0.20 | [-1.10, 0.40] | 0.50
```

```
.....
1 | 0 | 1 | 0 | 1 | [-1.10, 0.40] | 0.50 | [-0.40, 0.40] | 1.20
```

```
.....
1 | 1 | 0 | 1 | -1 | [-0.40, 0.40] | 1.20 | [-1.10, -0.30] | 0.50
```

```
.....
=====
Epoch : 6, learning_rate : 0.7
```

```
=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
```

```
-----
-> [Cycle Detected] Period: 4 updates
0 | 0 | 0 | 1 | -1 | [-1.10, -0.30] | 0.50 | [-1.10, -0.30] |
-0.20
```

```
.....
0 | 1 | 1 | 0 | 1 | [-1.10, -0.30] | -0.20 | [-1.10, 0.40] | 0.50
```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.10, 0.40] | 0.50 | [-0.40, 0.40] | 1.20
.....
1 | 1 | 0 | 1 | -1 | [-0.40, 0.40] | 1.20 | [-1.10, -0.30] | 0.50
.....

```

```

=====
Epoch : 7, learning_rate : 0.7
=====

```

```

X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [-1.10, -0.30] | 0.50 | [-1.10, -0.30] |
-0.20
.....
0 | 1 | 1 | 0 | 1 | [-1.10, -0.30] | -0.20 | [-1.10, 0.40] | 0.50
.....

```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.10, 0.40] | 0.50 | [-0.40, 0.40] | 1.20
.....
1 | 1 | 0 | 1 | -1 | [-0.40, 0.40] | 1.20 | [-1.10, -0.30] | 0.50
.....

```

```

=====
Epoch : 8, learning_rate : 0.7
=====

```

```

X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [-1.10, -0.30] | 0.50 | [-1.10, -0.30] |
-0.20
.....
0 | 1 | 1 | 0 | 1 | [-1.10, -0.30] | -0.20 | [-1.10, 0.40] | 0.50
.....

```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.10, 0.40] | 0.50 | [-0.40, 0.40] | 1.20
.....
1 | 1 | 0 | 1 | -1 | [-0.40, 0.40] | 1.20 | [-1.10, -0.30] | 0.50
.....

```

```

=====
Epoch : 9, learning_rate : 0.7
=====

```

```

X1 | X2 | Ac | Pred | Err | W_old          | b_old | W_new          | b_new
-----
0 | 0 | 0 | 1 | -1 | [-1.10, -0.30] | 0.50 | [-1.10, -0.30] |
-0.20
.....
0 | 1 | 1 | 0 | 1 | [-1.10, -0.30] | -0.20 | [-1.10, 0.40] | 0.50
.....

```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.10, 0.40] | 0.50 | [-0.40, 0.40] | 1.20
.....

```

```

.....
1 | 1 | 0 | 1 | -1 | [-0.40, 0.40] | 1.20 | [-1.10, -0.30] | 0.50

```

```

=====

```

Epoch : 10, learning_rate : 0.7

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----

```

```

0 | 0 | 0 | 1 | -1 | [-1.10, -0.30] | 0.50 | [-1.10, -0.30] |
-0.20

```

```

.....
0 | 1 | 1 | 0 | 1 | [-1.10, -0.30] | -0.20 | [-1.10, 0.40] | 0.50

```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.10, 0.40] | 0.50 | [-0.40, 0.40] | 1.20

```

```

.....
1 | 1 | 0 | 1 | -1 | [-0.40, 0.40] | 1.20 | [-1.10, -0.30] | 0.50

```

```

=====

```

Epoch : 1, learning_rate : 0.8

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----

```

```

0 | 0 | 0 | 1 | -1 | [1.00, -1.00] | 0.50 | [1.00, -1.00] | -0.30

```

```

.....
0 | 1 | 1 | 0 | 1 | [1.00, -1.00] | -0.30 | [1.00, -0.20] | 0.50

```

```

.....
1 | 0 | 1 | 1 | 0 | [1.00, -0.20] | 0.50 | [1.00, -0.20] | 0.50

```

```

.....
1 | 1 | 0 | 1 | -1 | [1.00, -0.20] | 0.50 | [0.20, -1.00] | -0.30

```

```

=====

```

Epoch : 2, learning_rate : 0.8

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----

```

```

0 | 0 | 0 | 0 | 0 | [0.20, -1.00] | -0.30 | [0.20, -1.00] | -0.30

```

```

.....
0 | 1 | 1 | 0 | 1 | [0.20, -1.00] | -0.30 | [0.20, -0.20] | 0.50

```

```

.....
1 | 0 | 1 | 1 | 0 | [0.20, -0.20] | 0.50 | [0.20, -0.20] | 0.50

```

```

.....
1 | 1 | 0 | 1 | -1 | [0.20, -0.20] | 0.50 | [-0.60, -1.00] | -0.30

```

```

=====

```

Epoch : 3, learning_rate : 0.8

```

=====

```

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[-0.60, -1.00]	-0.30	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

Epoch : 4, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.60, -1.00]	0.50	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

Epoch : 5, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.60, -1.00]	0.50	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

Epoch : 6, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.60, -1.00]	0.50	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

Epoch : 7, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.60, -1.00]	0.50	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

Epoch : 8, learning_rate : 0.8

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.60, -1.00]	0.50	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

=====
Epoch : 9, learning_rate : 0.8
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.60, -1.00]	0.50	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

=====
Epoch : 10, learning_rate : 0.8
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.60, -1.00]	0.50	[-0.60, -1.00]	-0.30
0	1	1	0	1	[-0.60, -1.00]	-0.30	[-0.60, -0.20]	0.50
1	0	1	0	1	[-0.60, -0.20]	0.50	[0.20, -0.20]	1.30
1	1	0	1	-1	[0.20, -0.20]	1.30	[-0.60, -1.00]	0.50

=====
Epoch : 1, learning_rate : 0.9
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.40
0	1	1	0	1	[1.00, -1.00]	-0.40	[1.00, -0.10]	0.50
1	0	1	1	0	[1.00, -0.10]	0.50	[1.00, -0.10]	0.50
1	1	0	1	-1	[1.00, -0.10]	0.50	[0.10, -1.00]	-0.40

=====
Epoch : 2, learning_rate : 0.9
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.10, -1.00]	-0.40	[0.10, -1.00]	-0.40
0	1	1	0	1	[0.10, -1.00]	-0.40	[0.10, -0.10]	0.50
1	0	1	1	0	[0.10, -0.10]	0.50	[0.10, -0.10]	0.50
1	1	0	1	-1	[0.10, -0.10]	0.50	[-0.80, -1.00]	-0.40

Epoch : 3, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[-0.80, -1.00]	-0.40	[-0.80, -1.00]	-0.40
0	1	1	0	1	[-0.80, -1.00]	-0.40	[-0.80, -0.10]	0.50
1	0	1	0	1	[-0.80, -0.10]	0.50	[0.10, -0.10]	1.40
1	1	0	1	-1	[0.10, -0.10]	1.40	[-0.80, -1.00]	0.50

Epoch : 4, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.80, -1.00]	0.50	[-0.80, -1.00]	-0.40
0	1	1	0	1	[-0.80, -1.00]	-0.40	[-0.80, -0.10]	0.50
1	0	1	0	1	[-0.80, -0.10]	0.50	[0.10, -0.10]	1.40
1	1	0	1	-1	[0.10, -0.10]	1.40	[-0.80, -1.00]	0.50

-> [Cycle Detected] Period: 4 updates

Epoch : 5, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
----	----	----	------	-----	-------	-------	-------	-------

```

-----
0 | 0 | 0 | 1 | -1 | [-0.80, -1.00] | 0.50 | [-0.80, -1.00] |
-0.40
.....
0 | 1 | 1 | 0 | 1 | [-0.80, -1.00] | -0.40 | [-0.80, -0.10] | 0.50
.....
1 | 0 | 1 | 0 | 1 | [-0.80, -0.10] | 0.50 | [0.10, -0.10] | 1.40
.....
1 | 1 | 0 | 1 | -1 | [0.10, -0.10] | 1.40 | [-0.80, -1.00] | 0.50
.....

```

```

=====
Epoch : 6, learning_rate : 0.9
=====

```

```

X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [-0.80, -1.00] | 0.50 | [-0.80, -1.00] |
-0.40
.....
0 | 1 | 1 | 0 | 1 | [-0.80, -1.00] | -0.40 | [-0.80, -0.10] | 0.50
.....
1 | 0 | 1 | 0 | 1 | [-0.80, -0.10] | 0.50 | [0.10, -0.10] | 1.40
.....
1 | 1 | 0 | 1 | -1 | [0.10, -0.10] | 1.40 | [-0.80, -1.00] | 0.50
.....

```

```

=====
Epoch : 7, learning_rate : 0.9
=====

```

```

X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [-0.80, -1.00] | 0.50 | [-0.80, -1.00] |
-0.40
.....
0 | 1 | 1 | 0 | 1 | [-0.80, -1.00] | -0.40 | [-0.80, -0.10] | 0.50
.....
1 | 0 | 1 | 0 | 1 | [-0.80, -0.10] | 0.50 | [0.10, -0.10] | 1.40
.....
1 | 1 | 0 | 1 | -1 | [0.10, -0.10] | 1.40 | [-0.80, -1.00] | 0.50
.....

```

```

=====
Epoch : 8, learning_rate : 0.9
=====

```

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.80, -1.00]	0.50	[-0.80, -1.00]	-0.40
0	1	1	0	1	[-0.80, -1.00]	-0.40	[-0.80, -0.10]	0.50
1	0	1	0	1	[-0.80, -0.10]	0.50	[0.10, -0.10]	1.40
1	1	0	1	-1	[0.10, -0.10]	1.40	[-0.80, -1.00]	0.50

Epoch : 9, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.80, -1.00]	0.50	[-0.80, -1.00]	-0.40
0	1	1	0	1	[-0.80, -1.00]	-0.40	[-0.80, -0.10]	0.50
1	0	1	0	1	[-0.80, -0.10]	0.50	[0.10, -0.10]	1.40
1	1	0	1	-1	[0.10, -0.10]	1.40	[-0.80, -1.00]	0.50

Epoch : 10, learning_rate : 0.9

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-0.80, -1.00]	0.50	[-0.80, -1.00]	-0.40
0	1	1	0	1	[-0.80, -1.00]	-0.40	[-0.80, -0.10]	0.50
1	0	1	0	1	[-0.80, -0.10]	0.50	[0.10, -0.10]	1.40
1	1	0	1	-1	[0.10, -0.10]	1.40	[-0.80, -1.00]	0.50

Epoch : 1, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[1.00, -1.00]	0.50	[1.00, -1.00]	-0.50
0	1	1	0	1	[1.00, -1.00]	-0.50	[1.00, 0.00]	0.50
1	0	1	1	0	[1.00, 0.00]	0.50	[1.00, 0.00]	0.50
1	1	0	1	-1	[1.00, 0.00]	0.50	[0.00, -1.00]	-0.50

Epoch : 2, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[0.00, -1.00]	-0.50	[0.00, -1.00]	-0.50
0	1	1	0	1	[0.00, -1.00]	-0.50	[0.00, 0.00]	0.50
1	0	1	1	0	[0.00, 0.00]	0.50	[0.00, 0.00]	0.50
1	1	0	1	-1	[0.00, 0.00]	0.50	[-1.00, -1.00]	-0.50

Epoch : 3, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	0	0	[-1.00, -1.00]	-0.50	[-1.00, -1.00]	-0.50
0	1	1	0	1	[-1.00, -1.00]	-0.50	[-1.00, 0.00]	0.50
1	0	1	0	1	[-1.00, 0.00]	0.50	[0.00, 0.00]	1.50
1	1	0	1	-1	[0.00, 0.00]	1.50	[-1.00, -1.00]	0.50

Epoch : 4, learning_rate : 1.0

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
-> [Cycle Detected] Period: 4 updates								
0	0	0	1	-1	[-1.00, -1.00]	0.50	[-1.00, -1.00]	-0.50
0	1	1	0	1	[-1.00, -1.00]	-0.50	[-1.00, 0.00]	0.50

```

.....
1 | 0 | 1 | 0 | 1 | [-1.00, 0.00] | 0.50 | [0.00, 0.00] | 1.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 1.50 | [-1.00, -1.00] | 0.50
.....

```

Epoch : 5, learning_rate : 1.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [-1.00, -1.00] | 0.50 | [-1.00, -1.00] |
-0.50
.....
0 | 1 | 1 | 0 | 1 | [-1.00, -1.00] | -0.50 | [-1.00, 0.00] | 0.50
.....

```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.00, 0.00] | 0.50 | [0.00, 0.00] | 1.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 1.50 | [-1.00, -1.00] | 0.50
.....

```

Epoch : 6, learning_rate : 1.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [-1.00, -1.00] | 0.50 | [-1.00, -1.00] |
-0.50
.....
0 | 1 | 1 | 0 | 1 | [-1.00, -1.00] | -0.50 | [-1.00, 0.00] | 0.50
.....

```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.00, 0.00] | 0.50 | [0.00, 0.00] | 1.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 1.50 | [-1.00, -1.00] | 0.50
.....

```

Epoch : 7, learning_rate : 1.0

```

=====
X1 | X2 | Ac | Pred | Err | W_old | b_old | W_new | b_new
-----
0 | 0 | 0 | 1 | -1 | [-1.00, -1.00] | 0.50 | [-1.00, -1.00] |
-0.50
.....
0 | 1 | 1 | 0 | 1 | [-1.00, -1.00] | -0.50 | [-1.00, 0.00] | 0.50
.....

```

```

.....
1 | 0 | 1 | 0 | 1 | [-1.00, 0.00] | 0.50 | [0.00, 0.00] | 1.50
.....
1 | 1 | 0 | 1 | -1 | [0.00, 0.00] | 1.50 | [-1.00, -1.00] | 0.50
.....

```

=====
Epoch : 8, learning_rate : 1.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-1.00, -1.00]	0.50	[-1.00, -1.00]	-0.50
0	1	1	0	1	[-1.00, -1.00]	-0.50	[-1.00, 0.00]	0.50
1	0	1	0	1	[-1.00, 0.00]	0.50	[0.00, 0.00]	1.50
1	1	0	1	-1	[0.00, 0.00]	1.50	[-1.00, -1.00]	0.50

=====
Epoch : 9, learning_rate : 1.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-1.00, -1.00]	0.50	[-1.00, -1.00]	-0.50
0	1	1	0	1	[-1.00, -1.00]	-0.50	[-1.00, 0.00]	0.50
1	0	1	0	1	[-1.00, 0.00]	0.50	[0.00, 0.00]	1.50
1	1	0	1	-1	[0.00, 0.00]	1.50	[-1.00, -1.00]	0.50

=====
Epoch : 10, learning_rate : 1.0
=====

X1	X2	Ac	Pred	Err	W_old	b_old	W_new	b_new
0	0	0	1	-1	[-1.00, -1.00]	0.50	[-1.00, -1.00]	-0.50
0	1	1	0	1	[-1.00, -1.00]	-0.50	[-1.00, 0.00]	0.50
1	0	1	0	1	[-1.00, 0.00]	0.50	[0.00, 0.00]	1.50
1	1	0	1	-1	[0.00, 0.00]	1.50	[-1.00, -1.00]	0.50

=====
XOR Gate Trace (End-of-Epoch Parameters & Cycle Info)
=====

[LR = 0.0] Total Updates: 30 | Cycle Detected: True (Period: 1)

End of Epoch 1: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 2: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 3: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 4: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 5: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 6: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 7: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 8: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 9: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 10: $w1 = 1.00$, $w2 = -1.00$, $b = 0.50$

[LR = 0.1] Total Updates: 26 | Cycle Detected: False (Period: 0)

End of Epoch 1: $w1 = 0.90$, $w2 = -1.00$, $b = 0.40$
End of Epoch 2: $w1 = 0.80$, $w2 = -1.00$, $b = 0.30$
End of Epoch 3: $w1 = 0.70$, $w2 = -1.00$, $b = 0.20$
End of Epoch 4: $w1 = 0.60$, $w2 = -1.00$, $b = 0.10$
End of Epoch 5: $w1 = 0.60$, $w2 = -0.90$, $b = 0.10$
End of Epoch 6: $w1 = 0.60$, $w2 = -0.80$, $b = 0.10$
End of Epoch 7: $w1 = 0.50$, $w2 = -0.80$, $b = 0.00$
End of Epoch 8: $w1 = 0.50$, $w2 = -0.70$, $b = 0.00$
End of Epoch 9: $w1 = 0.50$, $w2 = -0.60$, $b = 0.00$
End of Epoch 10: $w1 = 0.40$, $w2 = -0.60$, $b = -0.10$

[LR = 0.2] Total Updates: 23 | Cycle Detected: False (Period: 0)

End of Epoch 1: $w1 = 0.80$, $w2 = -1.00$, $b = 0.30$
End of Epoch 2: $w1 = 0.60$, $w2 = -1.00$, $b = 0.10$
End of Epoch 3: $w1 = 0.60$, $w2 = -0.80$, $b = 0.10$
End of Epoch 4: $w1 = 0.40$, $w2 = -0.80$, $b = -0.10$
End of Epoch 5: $w1 = 0.40$, $w2 = -0.60$, $b = 0.10$
End of Epoch 6: $w1 = 0.20$, $w2 = -0.60$, $b = -0.10$
End of Epoch 7: $w1 = 0.20$, $w2 = -0.40$, $b = 0.10$
End of Epoch 8: $w1 = 0.00$, $w2 = -0.40$, $b = -0.10$
End of Epoch 9: $w1 = 0.00$, $w2 = -0.20$, $b = 0.10$
End of Epoch 10: $w1 = -0.20$, $w2 = -0.20$, $b = -0.10$

[LR = 0.3] Total Updates: 25 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.70$, $w2 = -1.00$, $b = 0.20$
End of Epoch 2: $w1 = 0.40$, $w2 = -1.00$, $b = -0.10$
End of Epoch 3: $w1 = 0.40$, $w2 = -0.70$, $b = 0.20$
End of Epoch 4: $w1 = 0.10$, $w2 = -0.70$, $b = -0.10$
End of Epoch 5: $w1 = 0.10$, $w2 = -0.40$, $b = 0.20$
End of Epoch 6: $w1 = -0.20$, $w2 = -0.40$, $b = -0.10$
End of Epoch 7: $w1 = -0.20$, $w2 = -0.10$, $b = 0.20$
End of Epoch 8: $w1 = -0.50$, $w2 = -0.10$, $b = -0.10$
End of Epoch 9: $w1 = -0.50$, $w2 = -0.10$, $b = 0.20$
End of Epoch 10: $w1 = -0.50$, $w2 = -0.10$, $b = 0.20$

[LR = 0.4] Total Updates: 33 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.60$, $w2 = -1.00$, $b = 0.10$
End of Epoch 2: $w1 = 0.20$, $w2 = -1.00$, $b = -0.30$

End of Epoch 3: $w1 = 0.20$, $w2 = -0.60$, $b = 0.10$
End of Epoch 4: $w1 = -0.20$, $w2 = -0.60$, $b = -0.30$
End of Epoch 5: $w1 = -0.20$, $w2 = -0.60$, $b = 0.10$
End of Epoch 6: $w1 = -0.20$, $w2 = -0.60$, $b = 0.10$
End of Epoch 7: $w1 = -0.20$, $w2 = -0.60$, $b = 0.10$
End of Epoch 8: $w1 = -0.20$, $w2 = -0.60$, $b = 0.10$
End of Epoch 9: $w1 = -0.20$, $w2 = -0.60$, $b = 0.10$
End of Epoch 10: $w1 = -0.20$, $w2 = -0.60$, $b = 0.10$

[LR = 0.5] Total Updates: 33 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.50$, $w2 = -1.00$, $b = 0.00$
End of Epoch 2: $w1 = 0.00$, $w2 = -1.00$, $b = -0.50$
End of Epoch 3: $w1 = 0.00$, $w2 = -0.50$, $b = 0.00$
End of Epoch 4: $w1 = -0.50$, $w2 = -0.50$, $b = -0.50$
End of Epoch 5: $w1 = -0.50$, $w2 = -0.50$, $b = 0.00$
End of Epoch 6: $w1 = -0.50$, $w2 = -0.50$, $b = 0.00$
End of Epoch 7: $w1 = -0.50$, $w2 = -0.50$, $b = 0.00$
End of Epoch 8: $w1 = -0.50$, $w2 = -0.50$, $b = 0.00$
End of Epoch 9: $w1 = -0.50$, $w2 = -0.50$, $b = 0.00$
End of Epoch 10: $w1 = -0.50$, $w2 = -0.50$, $b = 0.00$

[LR = 0.6] Total Updates: 32 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.40$, $w2 = -1.00$, $b = -0.10$
End of Epoch 2: $w1 = -0.20$, $w2 = -1.00$, $b = -0.10$
End of Epoch 3: $w1 = -0.20$, $w2 = -0.40$, $b = 0.50$
End of Epoch 4: $w1 = -0.80$, $w2 = -0.40$, $b = -0.10$
End of Epoch 5: $w1 = -0.80$, $w2 = -0.40$, $b = 0.50$
End of Epoch 6: $w1 = -0.80$, $w2 = -0.40$, $b = 0.50$
End of Epoch 7: $w1 = -0.80$, $w2 = -0.40$, $b = 0.50$
End of Epoch 8: $w1 = -0.80$, $w2 = -0.40$, $b = 0.50$
End of Epoch 9: $w1 = -0.80$, $w2 = -0.40$, $b = 0.50$
End of Epoch 10: $w1 = -0.80$, $w2 = -0.40$, $b = 0.50$

[LR = 0.7] Total Updates: 32 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.30$, $w2 = -1.00$, $b = -0.20$
End of Epoch 2: $w1 = -0.40$, $w2 = -1.00$, $b = -0.20$
End of Epoch 3: $w1 = -0.40$, $w2 = -0.30$, $b = 0.50$
End of Epoch 4: $w1 = -1.10$, $w2 = -0.30$, $b = -0.20$
End of Epoch 5: $w1 = -1.10$, $w2 = -0.30$, $b = 0.50$
End of Epoch 6: $w1 = -1.10$, $w2 = -0.30$, $b = 0.50$
End of Epoch 7: $w1 = -1.10$, $w2 = -0.30$, $b = 0.50$
End of Epoch 8: $w1 = -1.10$, $w2 = -0.30$, $b = 0.50$
End of Epoch 9: $w1 = -1.10$, $w2 = -0.30$, $b = 0.50$
End of Epoch 10: $w1 = -1.10$, $w2 = -0.30$, $b = 0.50$

[LR = 0.8] Total Updates: 36 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.20$, $w2 = -1.00$, $b = -0.30$
End of Epoch 2: $w1 = -0.60$, $w2 = -1.00$, $b = -0.30$
End of Epoch 3: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$
End of Epoch 4: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$

End of Epoch 5: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$
End of Epoch 6: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$
End of Epoch 7: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$
End of Epoch 8: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$
End of Epoch 9: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$
End of Epoch 10: $w1 = -0.60$, $w2 = -1.00$, $b = 0.50$

[LR = 0.9] Total Updates: 36 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.10$, $w2 = -1.00$, $b = -0.40$
End of Epoch 2: $w1 = -0.80$, $w2 = -1.00$, $b = -0.40$
End of Epoch 3: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$
End of Epoch 4: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$
End of Epoch 5: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$
End of Epoch 6: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$
End of Epoch 7: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$
End of Epoch 8: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$
End of Epoch 9: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$
End of Epoch 10: $w1 = -0.80$, $w2 = -1.00$, $b = 0.50$

[LR = 1.0] Total Updates: 36 | Cycle Detected: True (Period: 4)

End of Epoch 1: $w1 = 0.00$, $w2 = -1.00$, $b = -0.50$
End of Epoch 2: $w1 = -1.00$, $w2 = -1.00$, $b = -0.50$
End of Epoch 3: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 4: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 5: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 6: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 7: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 8: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 9: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$
End of Epoch 10: $w1 = -1.00$, $w2 = -1.00$, $b = 0.50$